Critical sets: records and prolongation bounds

New Bunya records and verified product profiles; bounds through order 257 17 October 2026

Latest verified update. Full Bunya 30001580 records prove sizes 152, 462, 540, 872 and 3363 at orders 15, 27, 28, 33 and 65. Every clue-deletion witness has been replayed and two independent encodings exclude a second completion. Orders 15 and 65 additionally have complete, solver-free uniqueness trees. The export improves the earlier order-27 report from 461 to 462. See page 2. Verified profiles also strengthen lifted constructions; the order-65 seed gives $\mathrm{lcs}(130) \geq 14191$ (page 4).

Let $\alpha = \log_2 3$. The new independently verified critical set proves

$$\boxed{\mathrm{lcs}(33) \geq 872 > 834 = \lceil 33^2 - 33^\alpha \rceil.}$$

Replacing the original right side $E(n) = n^2 - n^\alpha$ by its ceiling **does not repair the upper conjecture**. This partial has 872 clues and 217 holes, whereas that conjecture requires at least 255 holes. Both the original real upper bound and the rounded version are false.

Construction and verification. The earlier verified Bunya record comes from array 29909858, task 0, rank 5. The attachment was cut after 6000 characters, but both grids are complete; the missing size and certification labels were not needed. It uses the same `lab_special33` completion and the archived 875-clue input, with 31 clues removed. Two encodings exclude alternate completions. A separate agent independently confirms uniqueness and replays all 844 essential-clue witnesses. Uniqueness remains solver-dependent; the witnesses are solver-free to replay. Details and exact grids are on pages 55 and 53.

Verified milestone. Bunya array 29909858 also gives $\mathrm{lcs}(15) \geq 151$, replacing 150. A complete uniqueness tree and all clue-deletion witnesses have been independently replayed (page 47). The verified size 152 now supersedes this milestone; both certificates are retained.

What remains. The earlier verified 841 is superseded. The current unconditional interval is $872 \leq \mathrm{lcs}(33) \leq 977$; no optimum is claimed. The latest independently verified order-27 record is 462 (page 2), fifteen above 447. The lower-side proposal remains

$$\max\{L_{\mathrm{proved}}(n), B(n), C(n)\} \overset{?}{\leq} \mathrm{lcs}(n),$$

where L_{proved} includes proved constructions, verified examples and protected lifts. Only B, C are conjectural terms. The graphs retain $U(n) = \lceil E(n) \rceil = n^2 - \lfloor n^\alpha \rfloor$ as a *refuted reference curve*, without shading it as an admissible band. No fitted coefficient or new ad hoc upper correction is substituted. The powers-of-two equality $\mathrm{lcs}(2^m) = 4^m - 3^m$ is not refuted by this order-33 example.

The original functions retained from BCC19.2 are

$$\begin{aligned} B(n) &= \max_{\substack{pq=n \\ p,q\geq 1}} \frac{n(3n-p-q-1)}{4}, \\ C(n) &= \#\{(i,j) : 0 \leq i,j < n,\ i \,\&\, j \neq 0\}, \\ E(n) &= n^2 - n^\alpha. \end{aligned}$$

The symbol & means binary bitwise AND: C counts pairs sharing a 1-bit. B is the product expression obtained from triangular seeds. It is already integer, since each factorization gives $n^2 - [p(p+1)/2][q(q+1)/2]$.

What rounding changes. The old real inequality $\mathrm{lcs}(n) \leq E(n)$ is equivalent to $\mathrm{lcs}(n) \leq \lfloor E(n) \rfloor$. Replacing its right side by $\lceil E(n) \rceil$ is a genuinely weaker conjecture. At order 17, $E(17) = 199.830691\ldots$, so the ceiling admits the verified size 200. At $n = 2^m$ it changes nothing, and the binary construction would force

$$\mathrm{lcs}(2^m) = 4^m - 3^m.$$

The earlier records and 131 stored construction rows fitted U, but those finite checks did not prove it. The latest verified 872 exceeds $U(33)$ by 38.

Earlier incompatibilities. The unrestricted product D would give $\mathrm{lcs}(68) \geq 3823 > U(68) = 3822$; the former sharp-plus target gives $838 > U(33) = 834$. Refuting U does not prove either lower family. Verified 872 surpasses 838, ruling out equality at 33. The order-68 evidence is on page 37.

Proved upper curve. The graphs separately show the independently agent-audited bounds

$$\mathrm{lcs}(n) \leq \begin{cases} n^2 - \lceil 7n/2 - 6 \rceil, & 8 \leq n \leq 13, \\ n^2 - \lceil 7n/2 - 4 \rceil, & n \geq 14. \end{cases}$$

Their complete arguments, including the one-hole-symbol case, remain in `lcs_review_8_33.tex` and the accompanying upper-proof notes. Human refereeing and literature-priority checks remain outstanding.

Bunya 30001580: independently verified records

The complete five-record export supplied on 17 October 2026 has now passed independent verification. In particular, its order-27 record has **462** clues, improving the earlier collector report of 461. These entries now appear as verified black points and enter the proved lower envelope.

n	Previous proved	Verified	Gain	$\lceil E(n) \rceil$	$U_p(n)$
15	151	152	1	152	176
27	447	462	15	544	638
28	536	540	4	588	690
33	844	872	28	834	977
65	3282	3363	81	3478	4001

The earlier 3282 is a construction guarantee. No optimum is asserted. The unchanged reported sizes at orders 14, 17 and 30 retain their previous verification evidence; this export contains no new grids for those orders.

Essentiality and uniqueness. We checked all five complete squares for Latinness, counted the retained clues and checked agreement of the partials with their completions. For each of the **5389 clues**, its supplied deletion witness was reconstructed and checked to be a Latin square that changes that clue and preserves every other retained clue. Two separately implemented models then excluded a second completion: an integer-cell model with row and column AllDifferent constraints, and a Boolean exact-cover model with cell, row-symbol and column-symbol constraints. Both returned INFEASIBLE for every record. Each used OR-Tools 9.14.6206 with one worker; neither used the search program's singleton-propagation code. These are two encodings of the same solver, not two different solver engines.

n	Clues checked	Holes	Private t_2	Portable uniqueness certificate
15	152	73	54	3833-node complete tree
27	462	267	105	Solver-dependent
28	540	244	330	Solver-dependent
33	872	217	293	Solver-dependent
65	3363	862	3240	8439-node complete tree

Every private-intercalate count was independently recomputed over all pairs of rows and columns. It counts clues admitting a private intercalate, not all intercalates. Safe $t_3 = 0$ is used unless separately established.

Why the portable certificates prove uniqueness. At each node the replay checker recomputes all legal assignments to the selected unfilled cell, missing row-symbol pair, or missing column-symbol pair. It requires exactly one child for every legal assignment, with no missing or duplicate branches. A dead leaf has no legal assignment; a completed leaf must equal the supplied square. Each accepted tree has exactly one completed leaf. Exhaustion of all branches proves uniqueness independently of a solver verdict. Combined with the explicit deletion witnesses, this proves criticality. Both trees were generated and checked by separate programs. The 20-second tree-generation attempts at orders 27, 28 and 33 did not finish; no portable uniqueness claim is made for those three.

Consequences, provenance and audit

At order 15, the verified 152 reaches the rounded reference value; 153 would exceed it. At order 33, $872-834=38$ and $872-838=34$, strengthening the already established counterexample to the rounded upper proposal. The conjectural lower targets at orders 27, 28 and 30 remain 508, 559 and 663, leaving gaps of 46, 19 and 9. At order 65, the bitwise lower target 3368 is five above the verified 3363. Since

$$E(65) = 3477.86395806245\ldots, \qquad \lceil E(65) \rceil = 3478,$$

the first integer exceeding $E(65)$ is 3478 and the first exceeding its ceiling is 3479: respectively 115 and 116 further clues are needed.

New lifted consequences. Substituting the verified profiles into the previously audited binary and ternary theorems gives

$$\begin{aligned} \mathrm{lcs}(99) &\geq 2 \cdot 33^2 + 5 \cdot 872 = 6538, \\ \mathrm{lcs}(130) &\geq 65^2 + 2 \cdot 3363 + 3240 = 14191, \\ \mathrm{lcs}(195) &\geq 2 \cdot 65^2 + 5 \cdot 3363 = 25265. \end{aligned}$$

These improve the preceding construction entries 6398, 10789 and 24860. The full envelope still maximizes over all earlier profiles: a larger critical set need not have a better private-trade profile. In particular, the new order-28 profile does not replace the earlier seed in every lift. No unrestricted product assertion is used.

Provenance. The preserved input is `lcs_bunya_30001580/verification/source_records.jsonl`. Its five lines correspond to orders 15, 27, 28, 33 and 65. The record metadata identifies job 30001580 as follows (all rank 0):

Order	Export line	Task	Iteration	Clues
15	1	16	5037	152
27	2	22	1305	462
28	3	21	358	540
33	4	12	307	872
65	5	0	608	3363

Within that verification directory, each `nN/` subdirectory contains `bunya_record.json`, `input.in`, the two uniqueness verdicts, the recomputed profile, and any available portable tree. The source hash and per-record hashes are recorded in `summary.json`; the original size-only collector table remains archived in `reported_records.json` in the parent directory. This preserves the earlier 461 report without confusing it with the newer verified 462.

Reproduction. From the verification directory run

```
python3 replay_record.py n65/bunya_record.json -check-unique
python3 check_portable_certificates.py n65
```

The second command uses only the Python standard library and checks both the exhaustive uniqueness tree and all supplied deletion witnesses. The first requires OR-Tools. The full proof volume includes all five new completion/partial pairs; the order-65 matrices are printed in numeric row panels so their complete entries remain readable.

Independent agent audit. A separate agent wrote its own witness replay and row/column-mask tree checker, and separately regenerated the uniqueness models. It confirmed all 5389 deletion witnesses, both complete trees (3833 and 8439 nodes, each with one solution), and all five t_2 counts using row-permutation two-cycle enumeration. Negative tests rejected a missing branch and a false solution leaf. Its fresh uniqueness models returned INFEASIBLE for all five records, and it checked the three lifted inequalities above. The audit is PASS; its scripts, hashes, detailed scope and limitations are in `verification/independent_audit/` below the same directory. This is independent program/AI-agent verification, not human refereeing. Uniqueness at orders 27, 28 and 33 still depends on CP-SAT; the portable checks at 15 and 65 do not.

Verified product profiles and new lower bounds

Separate from the running search table, the completed array 29909858 and local package tests supplied fully available critical sets with useful private-intercalate profiles. These inputs have been independently checked and may be used in the proved construction envelope.

For a single critical seed of order m, size s, and t clues having a private intercalate, the already audited repeated-binary-lift theorem gives

$$\begin{aligned} H_2(m;s,t) &= m^2 + 2s + t, \\ H_4(m;s,t) &= 7m^2 + 3s + 6t, \\ H_8(m;s,t) &= 37m^2 + 27t, \\ H_{16}(m;s,t) &= 175m^2 - 27s + 108t. \end{aligned}$$

These bound $\mathrm{lcs}(2m), \mathrm{lcs}(4m), \mathrm{lcs}(8m), \mathrm{lcs}(16m)$ respectively. The count t is the number of protected clues, not the number of all intercalates in the completed square. All parameters must come from one seed. In particular the negative coefficient of s in H_{16} forbids substituting an unrelated larger record size.

Derivation from the audited theorem. In the existing mixed-lift notation put $N = m^2$, $D = m^2 - s$, $u_2 = s - t$ and use the safe value $t_3 = 0$. One, two, three and four binary steps give deficit coefficient vectors

$$(0,3,1,0), \quad (0,9,6,0), \quad (0,27,27,0), \quad (0,81,108,0)$$

on (N, D, u_2, u_3). Subtracting each resulting deficit from $(2^k m)^2$ gives the four displayed formulas. This is substitution into the previously audited recurrence, not an unrestricted product assertion.

m	s	t	Origin and independently checked evidence
14	114	86	Completed Bunya array: 114 deletion witnesses
27	439	187	Completed Bunya array: 439 deletion witnesses
28	536	344	Completed Bunya array: 536 deletion witnesses
14	95	89	Local package test: 95 deletion witnesses
14	96	91	Final local launch test: 96 deletion witnesses

For every seed all supplied deletion alternatives were replayed as Latin squares preserving every other clue. A fresh integer-cell AllDifferent model excluded a second completion. The private-intercalate counts were recomputed; both local profiles also passed a separate enumeration of all pairs of rows and columns. Uniqueness remains solver-dependent. These are independent program checks, not a new separate-agent or human review.

Order	Previous proved	New proved	Gain	Seed and lift
54	1782	1794	12	$(27, 439, 187)$, H_2
56	2191	2230	39	$(14, 114, 86)$, H_4
112	9142	9709	567	$(14, 96, 91)$, H_8
224	38269	41536	3267	$(14, 96, 91)$, H_{16}

These comparisons use the project's previous construction table. The earlier local $(14, 95, 89)$ guarantees 9655 and 41347 at orders 112 and 224; these remain valid but are superseded. The $(28, 536, 344)$ lift at order 56 gives 2200, less than 2230. Every coherent profile is retained when maximizing over mixed and repeated lifts.

Reproduction. Inputs, verification paths and safe $t_3 = 0$ values are listed in `lcs_bunya_30001580/lift_profiles.json`. The completed-run checks are in `lcs_review_29909858/profile_verification.json`; the two local checks are in `lcs_hpc_65_192/validation/`. Within that package, `verify_profile.pyseeds/profile_n14_96_t91.json` replays the witnesses, reruns integer-cell uniqueness and independently recounts the intercalates. The full proof volume retains these five exact matrix pairs in its atlas.

Explicit lower-bound functions: complete catalogue

The following tables collect *all lower-bound families used or discussed in this report*, including the seed-dependent and recursive families. They are a catalogue of the present results, not a claim to exhaust the literature. Each function contributes zero outside its stated domain; maximize over the admissible parameters and available certified seeds. All entries in the following construction, product and counting tables are **proved**. Conjectural terms are listed separately on page 10.

1. Closed families and finite records.

Function	Orders/domain	Value	Source or restriction
$A(n)$	All $n \geq 1$	$n(n-1)/2$	Triangular critical set
$Q_2(n)$	$n = 2m$, $m \geq 1$	$(5m^2 - 3m)/2$	Donovan (1998), Thm. 3.5; $m = 1$ direct
$Q_4(n)$	$n = 4m$, $m \geq 1$	$(23m^2 - 9m)/2$	Bean thesis, Thm. 5.3.4
$Q_{16}(n)$	$n = 16m$, $m \geq 1$	$(377m^2 - 27m)/2$	Two safe binary lifts of the thesis seed
$T_9(n)$	$9 \mid n$	$(29n^2 - 27n)/54$	Cao–Du, Thm. 4.1
$P_2(n)$	$n = 2^k$, $k \geq 1$	$4^k - 3^k$	Stinson–van Rees; explicit XOR set
$P_3(n)$	$n = 3^k$, $k \geq 1$	$9^k - 6 \cdot 7^{k-1}$	Gower; size-three seed at order three
$P_3^{(0)}(n)$	$n = 3^k$, $k \geq 1$	$9^k - 7^k$	Size-two ternary seed; dominated by P_3
$M_-^{(0)}(n)$	$n = 2^k - 1$, $k \geq 2$	$4^k - 3^k - 2^{k+1} + 3$	Historical Fu–Fu–Liao; dominated by M_-
$M_-(n)$	$n = 2^k - 1$, $k \geq 2$	$4^k - 3^k - 2^{k+1} + 4$	Audited one-clue improvement
$J(n)$	$n = 2^a m$, $a \geq 1$	$n^2 - 3^a m^2$	Even cyclic seed, then $a - 1$ binary lifts
$E_0(n)$	$1 \leq n \leq 7$	$0, 1, 3, 7, 11, 18, 25$ in order	Exact small-order values
$R(n)$	Certified/published record orders	Largest available verified or published example size	Zero elsewhere; includes exact small orders

For J, the strongest admissible a is the largest exponent such that $2^a \mid n$. The P_2 row is retained even though it is a special case of J. Likewise E_0 is included in R, and $M_-^{(0)}$ is dominated by M_-. These redundancies make the catalogue explicit without changing its maximum. No formula in this table assumes that a general direct product preserves criticality.

Attribution. The Q_2 bound is Donovan (1998), Theorem 3.5, *Critical sets for families of Latin squares*, *Utilitas Mathematica* 53, 3–16. It is cited, not originated, in Bean's thesis. The Q_4 seed is Bean's theorem. Other primary sources are Bean's thesis, Chapters 4–5, Gower, Ars Combinatoria 55 (2000), 293–317, and Cao–Du, *Utilitas Mathematica* 62 (2002), 217–236.

Record status. R retains 47, 151, 447 and 844 at orders 9, 15, 27 and 33, respectively, and the other independently checked records in the record table. The older unverified 445 at order 27 is superseded. $M_-(15) = 147$ is a lower bound, not an equality: the verified record is 151. The later reports 152, 461, 540, 872 and 3363 at orders 15, 27, 28, 33 and 65 are marked S and do not yet enter R or its recursive product lifts. The independently checked profile updates are listed on page 4.

Explicit lower-bound functions: prolongations

Put $q = 2^k$, $k \geq 2$, $1 \leq a < k$, and $a_0 = \lfloor k/2 \rfloor$. Define

$$g_{k,a} = q^2 - 3^k - \frac{3q}{2} - 2^{a-1} - 2^{k-a-1}, \qquad \gamma_{k,a} = c(\tau p^{-1}) + c(p^{-1}) + c(\tau),$$

where τ is the explicit map in (12), $p = I + \tau$, and c counts cycles on nonzero vectors. These are finite computable permutations, and $\gamma_{k,a} \geq 3$. All terms for a split use the same map.

Function	Orders/domain	Value	Source or restriction
M_+^{unif}	$n = q + 1$	$g_{k,a_0} + 5$	Uniform projected trade cover
M_+^{cyc}	$n = q + 1$	$\max_{1 \leq a < k} (g_{k,a} + 2 + \gamma_{k,a})$	Exact-cycle refinement; includes the uniform value
$P_{4,+}$	$n = 4m + 1$ odd $m \geq 3$	$(21m^2 - 7m - 22)/2$	Thesis prolongation; full proof p. 19
$P_{16,+}$	$n = 16m + 1$ odd $m \geq 3$	$182m^2 - 18m - 44$	Explicit binary template; p. 20
$P_{3,+}^{\mathrm{unif}}$	$n = 3q + 1$	$7q^2 - 5 \cdot 3^k - 3q/2$ $- 2^{a_0 - 1} - 2^{k - a_0 - 1} + 1$	Uniform ternary transfer
$P_{3,+}^{\mathrm{cyc}}$	$n = 3q + 1$	$2q^2 + 4(q^2 - 3^k) - 4$ $+ \max_{1 \leq a < k} (g_{k,a} + 2 + \gamma_{k,a})$	Exact-cycle ternary transfer; p. 23
$\mathcal{T}_+$	$n = 3q + 1$; eligible (L, Q, T, b)	$2q^2 + 4s - 2h + b$	General transfer; hypothesis below is essential
K_{34}	$n = 34$	769	Audited conjugate-block protected set
K_{68}	$n = 68$	$34^2 + 2 \cdot 769 + 403 = 3097$	Safe binary lift of the same profile
K_{136}	$n = 136$	$7 \cdot 34^2 + 3 \cdot 769 + 6 \cdot 403 = 12817$	Safe two-step lift of that profile

General transfer hypothesis. Q must be a critical set of size s in a specified order-q square L, with transversal T and $h = |T \setminus Q|$. After prolongation, U_T prescribes transformed Q, all transversal positions and the complete new border. The input b must bound the size of *every defining subset of that particular* U_T. An arbitrary lower bound on $\mathrm{lcs}(q + 1)$ does not suffice. Thus the two $P_{3,+}$ rows are valid specializations of $\mathcal{T}_+$; unrelated record sizes are not substituted for b.

Which values are evaluated. The numerical report includes the stored audited M_+^{cyc} instances and their uniform fallback. It computes all splits for each applicable $P_{3,+}^{\mathrm{cyc}}$ order shown. At order 33 the exact-cycle guarantee is 735, while $R(33) = 844$. At 49 the ternary value 1366 is dominated by $P_{16,+}(49) = 1540$. The K profile guarantees a critical subset of size at least 769 with at least 403 private-intercalate clues. Its two displayed lifts have positive coefficients, so these lower counts are sufficient; no exact size is assumed for arbitrary mixed lifts.

Explicit lower-bound functions: three prolongations

These additional functions come from three disjoint transversals of the thesis square. Their full proof and independent audit details start on page 31. The oddness restriction on m is essential.

Function	Orders/domain	Value	Source or restriction
$P^{\mathrm{can}}_{4,+3}$	$n = 4m+3$, odd $m \geq 5$	$(21m^2 - 25m - 72)/2$	Canonical private rectangles; dominated below
$P_{4,+3}$	$n = 4m+3$, odd $m \geq 7$	$(21m^2 - 19m - 98)/2$	Flexible private rectangles and $2m+3$ cover
$P^{(5)}_{4,+3}$	$n = 23$	167	Exact protected count at $m = 5$
K_{CRT}	$n = 27, 43, \ldots, 203$	Explicit finite table on page 36	Independently replayed protected cores, plus two corner clues
$H_2[P_{4,+3}]$	$n = 8m+6$, odd $m \geq 7$	$44m^2 - 3m - 141$	Same seed's private-intercalate profile
$H_4[P_{4,+3}]$	$n = 16m+12$, odd $m \geq 7$	$(371m^2 + 183m - 792)/2$	Two safe binary lifts of that profile

The extracted order-$q = 4m+3$ critical set has size at least $S_m = (21m^2 - 19m - 98)/2$ and at least $T_m = 7m^2 - 8m - 52$ clues with private intercalates. The last two rows substitute these lower counts into $q^2 + 2s + t_2$ and $7q^2 + 3s + 6t_2$, which have positive coefficients. They do not assume an explicit critical set of *exactly* S_m clues, or justify substituting S_m for an unknown actual size in an arbitrary mixed-word expression with a negative size coefficient.

The general H_3 row on the next page also gives

$$\mathrm{lcs}(12m+9) \geq 2(4m+3)^2 + 5S_m = \frac{169m^2 + m - 454}{2} \qquad (m \geq 7 \text{ odd}),$$

and all recursive lifts of established sizes remain available. The numerical envelope includes that consequence through its recursive H_3 calculation. The canonical row is explicitly retained although the flexible family, together with its exceptional value at 23, dominates it throughout its domain. These functions do not assert a new lower bound at 11 or 19.

Explicit lower-bound functions: certified seed products

Use one coherent certified seed tuple (m, s, t_2, t_3): order m, actual critical-set size s, and lower counts t_2, t_3 of clues with private intercalate and order-three-subsquare witnesses. These count clues, not trades. Maximize each row over the available tuples satisfying its domain. Put $D = m^2 - s$, $u_2 = s - t_2$, $u_3 = s - t_3$.

Function	Orders/domain	Value for one eligible seed	Source or restriction
H_2	$n = 2m$	$m^2 + 2s + t_2$	Protected binary product
H_3	$n = 3m$	$2m^2 + 5s + 2t_3$	Protected ternary product
H_4	$n = 4m$	$7m^2 + 3s + 6t_2$	Two binary lifts
H_8	$n = 8m$	$37m^2 + 27t_2$	Three binary lifts
H_{16}	$n = 16m$	$175m^2 - 27s + 108t_2$	Four binary lifts; actual coherent size required
H_{24}	$n = 24m$	$333m^2 + 189t_2$	Mixed word 3222; explicit useful specialization
F_2	$n = 2^a m$, $a \geq 1$	$n^2 - 3^a D - \beta_a u_2$	Repeated binary lifts
F_3	$n = 3^a m$, $a \geq 1$	$n^2 - 7^a D - \chi_a u_3$	Repeated ternary lifts
F_{23}	$n = m\prod_i w_i$, $w_i \in \{2, 3\}$	$n^2 - A_w m^2 - B_w D - C_w u_2 - D_w u_3$	All finite mixed words; coefficients defined below
$\Psi_{3,2}$	$q = 3^a m$, $a \geq 1$, $n = 2q$	$q^2 + 2F_3(q) + 7^a t_2$	Ternary run followed by one binary lift
$\Psi_{3,4}$	$q = 3^a m$, $a \geq 1$, $n = 4q$	$7q^2 + 3F_3(q) + 6 \cdot 7^a t_2$	Ternary run followed by two binary lifts

The $F_3(q)$ in the last two rows is evaluated using the *same seed* as t_2. Its inherited private-intercalate count is at least $7^a t_2$. The specialized rows are retained even when a mixed-word calculation duplicates or improves them.

Coefficients for repeated lifts.

$$\beta_0 = 0, \quad \beta_1 = 1, \quad \beta_2 = 6, \quad \beta_a = 27\,4^{a-3}\ (a \geq 3), \qquad \chi_0 = 0, \quad \chi_{a+1} = 9\chi_a + 2\max(0, 7^a - \chi_a).$$

Coefficients for mixed words. List the factors of w in forward construction order. Start at $(A, B, C, D) = (0, 1, 0, 0)$ and process the word in reverse, using

$$\begin{aligned} \mathcal{B}(A, B, C, D) &= (4A + D,\ 3B,\ 4C + z,\ 3D), \\ \mathcal{T}(A, B, C, D) &= (9A + 2C,\ 7B,\ 7C,\ 9D + 2z), \\ z &= \max(0, B - C - D). \end{aligned}$$

The resulting tuple is (A_w, B_w, C_w, D_w). The D in this tuple is a coefficient, distinct from the seed deficit $m^2 - s$. These explicit rules cover every finite word, so the catalogue does not hide an unnamed "additional lifts" term.

Seed restrictions. The largest size and largest witness counts from different squares cannot be combined. Unknown witness counts may be set to zero. For F_2, F_3, F_{23}, use the actual seed size: its coefficient can be negative after expansion. The positive-size formulas H_2, H_3, H_4 also admit established lower sizes, as used in the next table.

Explicit lower-bound functions: recursion and counting

The recursive functions use already established values at *smaller* orders, with witness counts set to zero. They therefore introduce no circular assumption about lcs.

Function	Orders/domain	Value	Source or restriction
$G_2(n)$	$n = 2m$	$m^2 + 2L_{\text{proved}}(m)$	H_2, taking $t_2 = 0$
$G_3(n)$	$n = 3m$	$2m^2 + 5L_{\text{proved}}(m)$	H_3, taking $t_3 = 0$
$G_4(n)$	$n = 4m$	$7m^2 + 3L_{\text{proved}}(m)$	H_4, taking $t_2 = 0$
$K_{\text{HM}}(n)$	$n \geq 2$	$\max\{0, \lceil f_{\text{HM}}(n) \rceil\}$	Hatami–Mahmoodian, Theorem 1
$K_0(n; h)$	$0 \leq h \leq n^2$; $U(n,h) < L_0(n)$	$n^2 - h + 1$	Unamplified finite counting; certified strict inequality
$K_\Sigma(n; d, h)$	$0 \leq h \leq d \leq n^2$; condition below	$n^2 - d + 1$	Superset-amplified finite counting

Published all-order counting function. With natural logarithms,

$$f_{\text{HM}}(n) = n^2 \left(1 - \frac{2 + \ln 2}{\ln n}\right) + n \left(1 + \frac{\ln(8\pi)}{\ln n}\right) - \frac{\ln 2}{\ln n}.$$

See Hatami–Mahmoodian, *A lower bound for the size of the largest critical sets in Latin squares*, Theorem 1. Integrality permits the ceiling. This function improves the previous proved envelope at 15 orders from 199 to 251; orders 8–33 are unchanged.

The exact counting condition. Put $N = n^2$ and define

$$L_0(n) = \frac{(n!)^{2n}}{n^{n^2}}, \qquad F_n(a) = \prod_{t=a+1}^{n} (t!)^{1/t}.$$

Write $h = an + b$, $0 \leq b < n$, and set

$$U(n,h) = \binom{N}{h} (n!)^{h/n} \frac{F_n(a)^{n-b} F_n(a+1)^b}{(a!)^{n-b}((a+1)!)^b}.$$

At $h = N$, omit the factor with exponent $b = 0$, so $F_n(n+1)$ is not needed. The admissibility condition for K_Σ is

$$\boxed{L_0(n)\binom{d}{h} > U(n,h).}$$

K_0 is the case $d = h$, retained separately to show the original counting function. Maximize over certified pairs (d, h); a rounded numerical approximation without a certified positive margin is not sufficient. The balanced partial-square count and superset argument are proved in `lcs_bounds.md`, equations (16)–(19), and the earlier full review.

Finite evaluation versus asymptotics. Stored certificates give $K_\Sigma = 988, 4967, 23462$ at $n = 50, 100, 200$ (below constructions there), and 54939, 171402, 746306 at $n = 295, 500, 1000$. The proved maximum includes K_{HM} and stored certified pairs, separately from the construction curve; it does not exhaustively optimize all counting pairs. Asymptotically,

$$\text{lcs}(n) \geq n^2 - (2.668162063280737765\ldots + o(1))n^{5/3}.$$

It is *not* an additional finite-order function with the $o(1)$ omitted.

Seed coverage and the complete maximum

Coverage of the seed families. Let $\theta(m) = m(m+2)/8$ for even m and $\theta(m) = 0$ for odd m. The special seed profiles used for the repeated binary rows are

seed order	s	t_2
$2m$	$(5m^2 - 3m)/2$	$m^2 + 3\theta(m)$
$4m$	$(23m^2 - 9m)/2$	$7m^2 + 9\theta(m)$.

Thus the generator's separately labelled Donovan and thesis iterates are instances of F_2, rather than omitted lower-bound functions. The stored finite private-witness profiles likewise instantiate the H, F, Ψ rows.

With each seed or parameter family maximized over its admissible, established inputs, the full proved maximum in this report is

$$\begin{aligned} L_{\mathrm{proved}}(n) = \max\{&A, Q_2, Q_4, Q_{16}, T_9, P_2, P_3, P_3^{(0)}, M_-^{(0)}, M_-, J, E_0, R, \\ &M_+^{\mathrm{unif}}, M_+^{\mathrm{cyc}}, P_{4,+}, P_{16,+}, P_{3,+}^{\mathrm{unif}}, P_{3,+}^{\mathrm{cyc}}, \mathcal{T}_+, \\ &K_{34}, K_{68}, K_{136}, H_2, H_3, H_4, H_8, H_{16}, H_{24}, F_2, F_3, F_{23}, \\ &\Psi_{3,2}, \Psi_{3,4}, G_2, G_3, G_4, K_{\mathrm{HM}}, K_0, K_\Sigma\}(n). \end{aligned}$$

This specifies all the proved families and special cases in the catalogue; the numerical implementation uses the documented finite seed and cycle tables, the stated closed formulas, stored counting certificates, and the safe recursive closure. An unevaluated general transfer instance is not assigned a guessed numerical value.

The currently retained *conjectural* left-hand side is

$$\boxed{\max\{L_{\mathrm{proved}}(n), B(n), C(n)\} \overset{?}{\leq} \mathrm{lcs}(n).}$$

Only the additional B, C terms are conjectural. They are included in the orange curve; the proved construction curve and verified records retain their separate meanings.

Function	Orders/domain	Proposed lower value	Status in the displayed LHS
$B(n)$	All $n \geq 1$	$\max_{pq=n} \left(n^2 - \frac{p(p+1)q(q+1)}{4} \right)$	Conjectural; included
$C(n)$	All $n \geq 1$	$\#\{(i,j) : 0 \leq i, j < n,\ i \,\&\, j \neq 0\}$	Conjectural for general n; included
$D_{s,t}(n)$	$n = mp$ seed sizes s, t	$n^2 - (m^2 - s)(p^2 - t)$	Unrestricted product proposal; unproved, not included
$S_+(n)$	$n = 2^k + 1$, $k \geq 3$	$4^k - 3^k + 2^{k+1} - 7$	Former sharp-plus lower proposal; unproved in general, not included

$D_{s,t}$ is the numerical product proposal, not a theorem that its standard template is critical. Its proposed order-68 value 3823 remains uncertified. $S_+(33) = 838$ has been surpassed by the verified 844; that proves this one lower instance, not the all-k formula, and rules out equality at 33. At powers of two, $C = P_2$ is proved; its general-order lower claim is the remaining conjectural part. The dyadic equality proposal adds an upper claim and supplies no additional LHS function beyond P_2.

The old $E(n)$ and $\lceil E(n) \rceil$ are upper proposals, not LHS terms; both are refuted by the order-33 record. The catalogue does not restore either as an upper bound. The audited proved upper curve remains separate.

Reproducibility. Function domains and witness restrictions are in these tables; the detailed proofs and independent audits follow in this document or the cited earlier review. Numeric sources are recorded in `lcs_dyadic_conjectures_data.json`; the generator uses exact powers at dyadic orders and 80-digit arithmetic for the refuted reference curve.

Latest records and revised targets

The table incorporates the latest reported records, with verification status shown separately. "Construction" omits a bare record at the destination, but allows verified smaller seeds in lifts. L is $\max\{L_{\text{proved}}, B, C\}$, $U = \lceil E \rceil$ is the refuted reference curve, and U_p is the audited proved upper. The row at 33 gives the verified $872 > U = 834$, exceeding the rounded reference by 38. "Gap" is $\max(0, L - \text{record})$, not distance to a proved optimum.

Order	Construction	Record	Status	L	U	U_p	Gap
8	37	37	P	37	37	42	0
9	39	47	V	47	49	55	0
10	55	59	V	59	62	71	0
11	55	74	V	74	77	88	0
12	90	90	V	90	93	108	0
13	78	107	V	107	111	129	0
14	112	126	V	126	131	151	0
15	147	152	V	152	152	176	0
16	175	175	V	175	175	204	0
17	158	200	V	200	200	233	0
18	189	224	V	224	227	265	0
19	171	245	V	245	255	298	0
20	265	281	V	281	285	334	0
21	234	304	V	304	317	371	0
22	299	340	V	340	350	411	0
23	253	367	V	367	386	452	0
24	405	405	V	405	422	496	0
25	300	449	V	449	461	541	0
26	415	482	V	482	502	589	0
27	435	462	V	508	544	638	46
28	532	540	V	559	588	690	19
29	479	624	V	624	634	743	0
30	651	654	V	663	681	799	9
31	721	721	V	721	730	856	0
32	781	781	V	781	781	916	0
33	735	872	V	872	834	977	0
65	3282	3363	V	3368	3478	4001	5

V: independently verified example or proved construction. **P**: published record. **R**: historical reported record without independent re-verification. **S**: search-certified, pending independent verification. Only independently verified or proved sizes enter the proved envelope; a matching size alone does not verify a grid.

The remaining lower targets at 27, 28 and 30 are respectively 508, 559 and 663, coming from C. At order 33, verified 872 surpasses both the former lower target 782 and the former upper 834. The ceiling upper conjecture is refuted. The sharp-plus lower target 838 is surpassed at order 33; the lower formula for every $2^m + 1$ remains unproved, and equality is false at 33. At order 27 the verified 462 is 46 below the target. At order 23 the verified lower bound is 367; the old upper reference is 386, while the proved upper bound is 452.

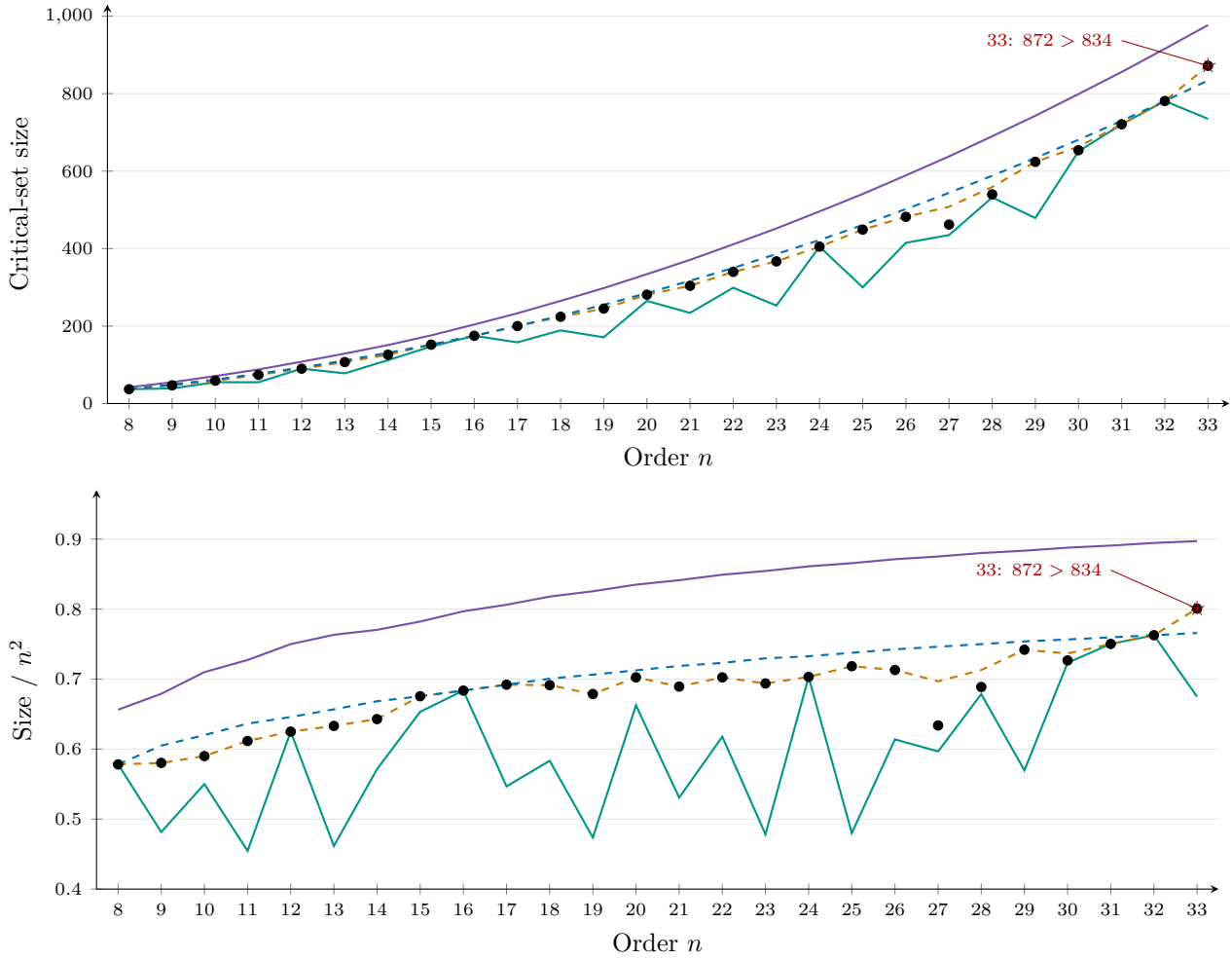
The earlier order-19 size-218 construction is retained on pages 42–43; the later size 245 has its own certificate. The current order-33 verification is on page 2; the full new grids are in the companion atlas. Earlier certificates are on pages 50 and 51; the historical order-15 and ten-record Bunya audits follow on pages 44 and 45.

Reported and verified records: orders 8–33

Solid teal: proved constructions. Solid purple: audited proved upper.

Dashed orange: conjectural lower. Dashed blue: refuted upper candidate $\lceil E(n) \rceil$.

Black circles: verified/published records. Open magenta triangles: search-certified, pending independent verification.



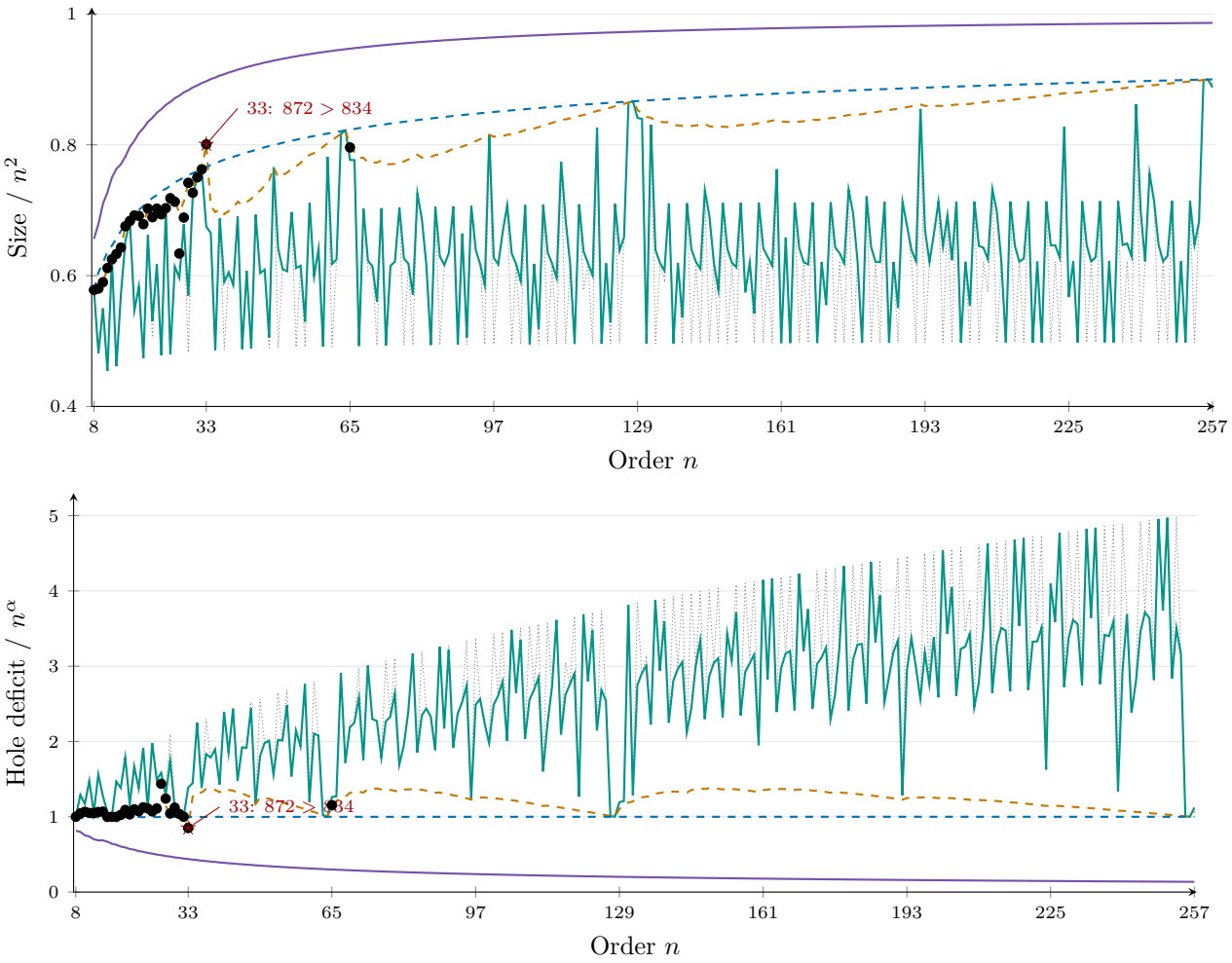
No feasible band is shaded: the verified point $(33, 872)$ exceeds the dashed blue value 834. A record below the lower candidate still indicates an open construction target. Lines join integer orders; construction fluctuations reflect arithmetic restrictions on the available families.

Larger-order view: orders 8–257

Solid teal: proved constructions. Solid purple: audited proved upper.

Dashed orange: conjectural lower. Dashed blue: refuted upper candidate $\lceil E(n) \rceil$.

Black circles: verified/published records. Open magenta triangles: search-certified, pending independent verification. Dotted grey: the previous construction envelope, using the same records.

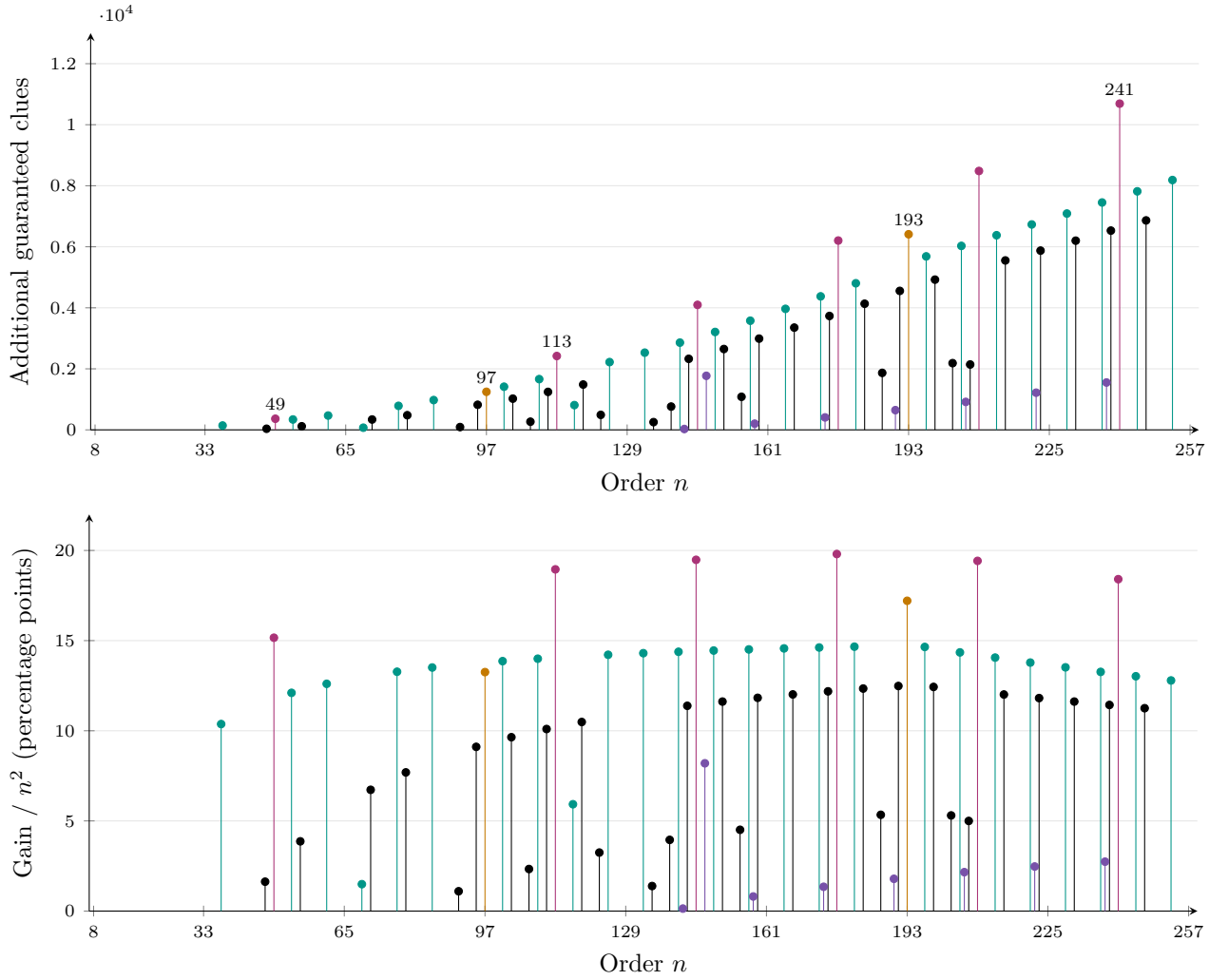


The second graph reverses the size ordering: fewer holes mean more clues. The constructive envelope is uneven because divisibility determines which families apply. The displayed candidate curves still meet at powers of two, but the global upper conjecture is false. This does not refute exactness specifically at powers of two. The surpassed target 838 is not drawn as a separate bound.

Where the new prolongations improve the bounds

The baseline uses the same verified records, older formulas and counting bounds, with all new prolongation families disabled throughout the recursive lifts. The new families improve the construction envelope at **73 orders** up to 257, and the overall proved lower bound at **71 orders**. At 21 and 29 the construction improves, but the existing records remain stronger.

Teal: $4m + 1$. Magenta: $16m + 1$. Orange: $3 \cdot 2^k + 1$. Black: $4m + 3$. Purple: safe product consequence.



Each stem is the increase in the overall proved lower bound at that integer order; no interpolation or positive gain is asserted at missing orders. The lower panel divides by n^2, so its units are percentage points of density.

Prolongation lower-bound functions

These are existence guarantees obtained by prolonging an explicit defining set and taking an inclusion-minimal defining subset. The starting defining set need not itself be critical. Each function contributes zero outside its stated domain. The three earlier families are:

$$\begin{aligned} P_{4,+}(4m+1) &= \frac{21m^2-7m-22}{2} && (m \geq 3 \text{ odd}), \\ P_{16,+}(16m+1) &= 182m^2-18m-44 && (m \geq 3 \text{ odd}), \\ P^{\text{unif}}_{3,+}(3\cdot 2^k+1) &= 7\cdot 4^k - 5\cdot 3^k - 3\cdot 2^{k-1} \\ &\quad - 2^{a-1} - 2^{k-a-1} + 1 && (k \geq 2,\ a = \lfloor k/2 \rfloor). \end{aligned}$$

The new $P_{4,+3}$ family and its safe profile lifts are listed on page 7 and proved on page 31. All these functions enter the maximum defining L_{proved}. For $P_{3,+}$ the numerical envelope maximizes the stronger exact-cycle formula over all splits $1 \leq a < k$, proved on page 23. The $4m+1$ result from the preceding investigation is included here for the first time. The full $4m+1$ and $16m+1$ proofs start on page 19; the new ternary transfer and its XOR specialization follow on page 23. Their independent audits are stated with their precise scopes alongside the proofs.

Order	Previous construction	New construction	Overall gain	Source
21	223	234	0	$4m+1$
29	406	479	0	$4m+1$
37	666	808	142	$4m+1$
49	1176	1540	364	$16m+1$
81	4503	4503	0	Existing maximum
97	4656	5903	1247	$3\cdot 2^k+1$
113	6328	8748	2420	$16m+1$
145	10440	14536	4096	$16m+1$
147	10731	12502	1771	Safe product
193	18528	24938	6410	$3\cdot 2^k+1$
241	28920	40636	10691	$16m+1$

At order 81 the $16m+1$ family gives 4416, below the existing 4503; the maximum retains 4503. At orders 13 and 25 the ternary family gives 60 and 299, also below the previous envelope. A transversal alone does not justify a critical-set lower bound, and no known record is substituted into a theorem that requires every defining subset of a specified partial square to have a certain size.

Propagation. The existing safe formulas may use a new established size with private-witness counts set to zero. For example, the new 1540 at order 49 gives $\text{lcs}(147) \geq 2\cdot 49^2 + 5\cdot 1540 = 12502$. This uses the positive-coefficient protected ternary lift; it makes no unrestricted product assumption. All 73 changed construction rows and their sources are exported in `lcs_prolongation_improvements.csv`.

Prolonging the thesis constructions: shared seed

The two constructions proved below give, for every **odd integer** $m \geq 3$,

$$\boxed{\operatorname{lcs}(4m+1) \geq \frac{21m^2 - 7m - 22}{2}}, \qquad \boxed{\operatorname{lcs}(16m+1) \geq 182m^2 - 18m - 44}. \tag{1}$$

Both start from an explicit defining set, preserve a specified collection of private trades under prolongation, and then take an inclusion-minimal defining subset. The sizes of the augmented defining sets themselves are not used as critical-set lower bounds. The common coordinate lemmas are given in full here.

The order-$4m$ square. Put $p = 4m$. Rows are (R, i) and columns (C, j), with $R, C \in \{0, 1, 2, 3\}$ and $0 \leq i, j < m$. The symbol block is $R \oplus C$, where $\oplus$ is bitwise XOR. Within a block, residues are modulo m. Write $A(i, j) = i - j$, $A^{\mathsf{T}}(i, j) = j - i$ and $B(i, j) = i + j$. The complete square L and clue mask of the thesis partial square P are

	0	1	2	3
0	A	A^{T}	A^{T}	B
1	A	A^{T}	B	A^{T}
2	A	B	A^{T}	A^{T}
3	B	A	A	A

	0	1	2	3
0	H	H	H	G
1	J	F	G	F
2	J	G	F	F
3	G	F	F	F

.

Here F prescribes every entry of the block; the other masks prescribe

$$H:\ 1 \leq j \leq i \leq m-1, \qquad J:\ 1 \leq i \leq j \leq m-1, \qquad G:\ i + j \geq m.$$

Each complete block is Latin, and its symbol block varies bijectively across each block row and column, so L is Latin. The established triangular critical-set construction, used in the thesis, says that each displayed H, J or G partial block is critical within its complete m-square. Thus every triangular clue has a private trade wholly within that block. This standard seed fact is an input; the product and prolongation arguments below do not assume a general critical-product theorem.

Unique completion of P. The full blocks in block row 3 force block 30 to use only symbol block 3; the full blocks in block column 3 do the same for block 03. Their G partials therefore complete uniquely to their prescribed copies of B. In block 12, its full row blocks $11, 13$ exclude symbol blocks $0, 2$, and its full column blocks $22, 32$ exclude $0, 1$. Only symbol block 3 remains. Similarly block 21 is restricted to symbol block 3. Their G partials complete uniquely. Now the other three blocks in each relevant row or column force blocks $10, 20, 01, 02$ into symbol blocks $1, 2, 1, 2$, respectively. Their J or H partials complete uniquely. Finally block 00 is restricted to symbol block 0 and its H partial completes. Hence P defines L.

There are seven full blocks and nine triangles, giving

$$s = |P| = 7m^2 + 9\frac{m(m-1)}{2} = \frac{23m^2 - 9m}{2}. \tag{2}$$

The private intercalates constructed on the next pages prove that every full-block clue is essential as well. Together with the triangular private trades, this recovers the criticality needed from the thesis seed.

Transversal, contraction and private rectangles

A shifted thesis transversal. Let t be the following column permutation of the inner order-$4m$ square:

Row	Column	Symbol block and residue
$(0,i)$	$(0,m-1-i)$	$(0,2i+1)$
$(1,i)$	$(2,i)$	$(3,2i)$
$(2,i)$	$(3,-i)$	$(1,-2i)$
$(3,i)$	$(1,-i)$	$(2,2i)$

All residues are modulo m. Rows and columns occur once each; the four symbol blocks differ, and multiplication by 2 is a permutation because m is odd. Thus t is a transversal. It meets triangular blocks only in 00 and 12, in $(m-1)/2$ clues each, and meets full blocks $23, 31$ in m clues each. Therefore

$$|t \cap P| = 3m-1, \qquad |t \setminus P| = m+1. \tag{3}$$

Safe prolongation and minimal pruning. For any Latin square M with transversal T, replace its transversal entries by a new symbol ∞, move each displaced symbol into its row's new column and its column's new row, and put ∞ in the new corner. If Q defines M, prescribe its transformed entries, all of T, and the entire new row and column; call this partial square $U(Q,T)$. In any completion, the fixed new border and the fixed ∞ transversal permit contraction: delete the border and restore its known displaced symbols. The result is a Latin completion of Q, so $U(Q,T)$ is defining.

A private trade for a clue of Q which avoids T survives unchanged in $U(Q,T)$: it meets neither the added transversal clues nor the new border. Call such a clue *protected*. Every defining subset of $U(Q,T)$ must contain it, since otherwise switching that trade gives a second completion. An inclusion-minimal defining subset exists because $U(Q,T)$ is finite; it is a critical set and contains every protected clue.

Private rectangles in the seven full inner blocks. For target $(R,i;C,j)$, take other row (S,x) and other column (D,y). The four positions, in order, are target, horizontal, vertical and diagonal:

$$(R,i;C,j), \quad (R,i;D,y), \quad (S,x;C,j), \quad (S,x;D,y).$$

Target block (R,C)	Other block coordinates (S,D)	Residue relation
$11, 22$	$(0,0)$	$x \equiv j-i+y \pmod m$
33	$(0,0)$	$x \equiv i-j+y \pmod m$
13	$(0,2)$	$x \equiv i-j+y \pmod m$
23	$(0,1)$	$x \equiv i-j+y \pmod m$
31	$(2,0)$	$x \equiv i-j+y \pmod m$
32	$(1,0)$	$x \equiv i-j+y \pmod m$

Opposite symbols agree under these relations; the two symbol blocks differ since $S \neq R$. Thus each rectangle is an intercalate.

For blocks $11, 22$, use $(x,y) = (0,0)$ if $i = j$; otherwise use $((j-i) \bmod m, 0)$ if $j = 0$ or $0 < i < j$, and $(0, (i-j) \bmod m)$ if $i = 0$ or $0 < j < i$. For every other full block use $(i-j, 0)$ if $i \geq j$ and $(0, j-i)$ if $i < j$. These are private witnesses for *all* full-block clues. In $11, 22$, the two zero-index cells are holes and the third is a hole by $j > x$ or $i > y$, with zero target indices covered by the masks. For the other five blocks the two G sums are i or j, below m, and the remaining H or J cell is in its zero row or column.

Which private witnesses avoid the transversal?

For later products it matters which corner of an inner witness must avoid t. The following counts describe *explicit sufficient sets* of full-block clues, without claiming that all possible witnesses were exhausted:

Required avoidance	Number of selected full-block clues
Target only	$7m^2 - 2m$
Horizontal corner only	$7m^2 - 2$
Vertical corner only	$7m^2 - 2$
Diagonal corner only	$7m^2 - 6$
All four corners	$7m^2 - 2m - 10$

Target. Among full blocks, t meets exactly the m cells $i + j \equiv 0 \pmod m$ in each of 23 and 31. Omit those targets and use any of the preceding unconstrained private rectangles.

Horizontal corner. The horizontal other-position can lie in a block containing t only for target block 13, when it lies in 12 and its transversal condition is $y = i$. In block 13 omit

$$(i, j) = (0, 0), \quad \big((m-1)/2, m-1\big),$$

and choose

$$(x, y) = \begin{cases} (i - j, 0), & i \geq j, \\ (0, j - i), & i < j,\ j \neq 2i, \\ (1, i + 1), & i < j,\ j = 2i < m - 1. \end{cases}$$

The first case has $y = i$ only at the omitted $(0, 0)$; the second avoids it by $j \neq 2i$; the last has $y = i + 1$. In the last case the H corner is a hole since $y > x$, and both G sums equal $j + 1 < m$. The only missing endpoint is $j = 2i = m - 1$, exactly the second excluded target. All other full blocks retain their unconstrained witness, losing only these two clues.

Vertical corner. The corresponding restriction occurs only for target block 32, whose vertical other-position is in 12 with condition $x = j$. Omit

$$(i, j) = (0, 0), \quad \big(m-1, (m-1)/2\big),$$

and choose $(x, y) = (0, j - i)$ if $i \leq j$, $(i - j, 0)$ if $i > j$ and $i \neq 2j$, and $(j + 1, 1)$ if $i > j$ and $i = 2j < m - 1$. The modified case has $x > y$, so its J corner is a hole; both G sums are $i + 1 < m$, and $x \neq j$. The two exclusions are the zero case and the sole unhandled endpoint. This again loses two selected clues.

Diagonal corner. Only targets in $11, 22, 33$ have the opposite corner in a block containing t, namely 00, where the condition is $x + y \equiv m - 1 \pmod m$. The unconstrained witnesses in $11, 22$ meet it only at target coordinates $(0, 1), (1, 0)$; in 33 only at $(0, m - 1), (m - 1, 0)$. For positive unequal indices the former nonzero coordinate is $|i - j| \leq m - 2$; a zero index gives $m - j$ or $m - i$ and precisely the listed exceptions. In 33 the sum is $|i - j|$. Thus six clues are omitted.

All four corners. The exclusions just described lie in different full blocks: two each in $11, 22, 33, 13, 32$, and m each in $23, 31$. The modified witnesses have no other constrained corner. Hence all four positions avoid t for $7m^2 - 2m - 10$ selected full-block clues. Adding the seven unaffected triangular blocks gives a protected inner set F_0, disjoint from blocks $00, 12$ and from the new border, of size

$$f = |F_0| = 7\frac{m(m-1)}{2} + 7m^2 - 2m - 10 = \frac{21m^2 - 11m - 20}{2}. \tag{4}$$

Order $4m+1$: the complete proof

The additional intercalate families. Prolong L along t and use $U(P,t)$. Extend each of the two omitted finite blocks 00 and 12 by the new row and column. Each extension is a Latin subsquare of order $m+1$: its restricted transversal contains all old symbols of that block, which move into its border, and the subsquare gains ∞. The two extensions have different finite row blocks and different finite column blocks, so intersect only at the new corner. Neither contains a clue of F_0.

In extended block 00, label a finite diagonal position by row i, paired with column $m-1-i$, and label the new corner by $*$. All $m+1$ diagonal positions contain ∞. For distinct finite labels i,j, the crossing cells have equal old residue $i+j+1 \pmod m$. For the pair $\{i,*\}$ they have the displaced residue $2i+1$. In extended block 12 the finite pairing is row i with column i; the corresponding residues are $i+j$ and $2i$ in symbol block 3. Thus every pair of diagonal positions is the opposite pair of an intercalate, whose other repeated value is an old symbol.

Complete-graph cover lemma. These intercalates form a K_{m+1} family: diagonal clues are vertices, and each off-diagonal cell belongs to exactly one pair-intercalate. Suppose a defining set selects k diagonal clues in one such subsquare. Each pair among the other $m+1-k$ vertices must be met by an off-diagonal clue, since switching an unmet intercalate gives another completion. Distinct pairs require distinct off-diagonal clues. Consequently

$$\left|C \cap S\right| \geq k + \binom{m+1-k}{2} \geq m. \tag{5}$$

Indeed, on putting $q = m+1-k$, the difference from m is $(q-1)(q-2)/2 \geq 0$ for every integer $0 \leq q \leq m+1$. The argument applies to any defining set, including one restricted to $U(P,t)$; restrictions on available clues cannot weaken it.

Critical subset and count. Choose an inclusion-minimal defining subset $C \subseteq U(P,t)$. Its surviving private trades force $F_0 \subseteq C$. Each of the two extended subsquares requires at least m further clues, by (5). Their only common cell is the new corner, so together they require at least $2m-1$ clues outside F_0. Therefore

$$\begin{aligned} |C| &\geq f + 2m - 1 = \frac{21m^2 - 11m - 20}{2} + 2m - 1 \\ &= \frac{21m^2 - 7m - 22}{2}, \end{aligned} \tag{6}$$

proving the first bound in (1). This includes $m=3$: the core has 68 clues and the two covers force five more. Equivalently,

$$\operatorname{lcs}(n) \geq \frac{21n^2 - 70n - 303}{32} \qquad (n \geq 13,\ n \equiv 5 \pmod 8).$$

Initial values and interpretation.

m	Output order $4m+1$	Protected core f	Proved family bound
3	13	68	73
5	21	225	234
7	29	466	479
9	37	791	808

These are guaranteed sizes of some critical subsets, not exact values of lcs or asserted records. Other verified examples can be stronger. The thesis's original prolongation discussion counted intercalates; the protected-clue and cover argument here supplies the critical-set bound.

Order $16m + 1$: an explicit defining template

Two binary layers without an assumed critical product. Keep $p = 4m$, and form $M = C_2^2 \times L$ of order $4p = 16m$: rows (a, r), columns (b, c), and symbols

$$M((a, r), (b, c)) = (a \oplus b, L(r, c)), \qquad 0 \le a, b < 4.$$

Define W by filling the seven outer blocks with $a \,\&\, b \neq 0$ completely and putting a copy of P in each of the other nine blocks. The symbol & means bitwise AND.

Why W is defining. Split the outer row and column labels by their high bit. The bottom-right binary quadrant is full, so it forces the high-bit symbol classes in the top-right and bottom-left quadrants. Each of these is itself a binary template: its bottom-right sub-block is full and the other three are copies of P. That full sub-block first fixes the low-bit symbol classes of its two adjacent sub-blocks. These complete uniquely from their copies of P, and then fix the class of the remaining sub-block, which also completes uniquely. After the top-right and bottom-left quadrants have completed, they force the high-bit class of the top-left quadrant. The same inner argument completes it. Thus W uniquely defines M; criticality of this unpruned template has not been assumed.

The product transversal. Take the affine outer mapping

$$\tau = (1, 3, 2, 0), \qquad (a \oplus \tau(a))_{a=0}^{3} = (1, 2, 0, 3).$$

Both lists are permutations, so

$$T : (a, r) \longmapsto (\tau(a), t(r))$$

is a transversal of M. It meets the partial outer blocks $01, 30$ and the full outer blocks $13, 22$. Prolong along T and use the defining set $U(W, T)$ from page 17.

Protected clues in the nine partial blocks. Seven copies of P are completely disjoint from T, and retain all s of their private-trade clues. Each affected partial block $01, 30$ retains the f clues of F_0 whose inner trades avoid t. Their total is

$$7s + 2f = \frac{203m^2 - 85m - 40}{2}. \tag{7}$$

Private outer rectangles. Let $K = \{(a, b) : a \,\&\, b \neq 0\}$. Each of its seven cells has a rectangle meeting K only at that target. The following table specifies the other outer row A and column B, and the rectangle's unique intersection with τ:

Outer target	(A, B)	Transversal corner and position
11	$(0, 0)$	vertical, 01
13	$(0, 2)$	target, 13
22	$(0, 0)$	target, 22
23	$(0, 1)$	diagonal, 01
31	$(2, 0)$	horizontal, 30
32	$(1, 0)$	horizontal, 30
33	$(0, 0)$	horizontal, 30

Direct XOR substitution verifies each intercalate and its sole full-block position. The table also verifies the sole transversal intersection. In general a transversal meets an intercalate in at most one cell: adjacent cells share a row or column, and opposite cells share a symbol.

Order $16m+1$: protected count and pruning

Inner holes in full outer blocks. For an inner hole (r,c) of P outside t, use its four identical inner-coordinate copies at the outer rectangle's positions. Only the target outer block is full, so this is a private product intercalate. It avoids T because $(r,c) \notin t$. Every full outer block retains

$$d_0 = p^2 - s - (m+1) = \frac{9m^2+7m-2}{2} \tag{8}$$

such clues, using (3).

Full inner-block clues in full outer blocks. Let a private inner intercalate for target (r,c) have other row r' and column c'. Combine it with the target outer rectangle (a,b), other row A and other column B, to obtain

$$\begin{array}{ll} ((a,r),(b,c)), & ((a,r),(B,c')), \\ ((A,r'),(b,c)), & ((A,r'),(B,c')). \end{array}$$

Opposite product symbols agree. The other three outer blocks are partial, and their inner positions are holes of P, so this is private in W. Exactly one outer position lies on τ; hence it avoids T precisely when the corresponding inner corner avoids t. Applying the counts on page 18 gives

Outer block(s)	Inner condition	Selected clues per block
$11, 31, 32, 33$	vertical or horizontal	$d_0 + 7m^2 - 2$
$13, 22$	target	$d_0 + 7m^2 - 2m$
23	diagonal	$d_0 + 7m^2 - 6$

The inner-hole class and full-inner-clue class are disjoint. Summing the seven full blocks and adding (7) gives

$$\begin{aligned} \text{full-block total} &= 7d_0 + 49m^2 - 4m - 14 = \frac{161m^2+41m-42}{2}, \\ |F_{16}| &= \frac{203m^2-85m-40}{2} + \frac{161m^2+41m-42}{2} = 182m^2 - 22m - 41. \end{aligned} \tag{9}$$

Every defining subset of $U(W,T)$ contains this explicit protected core.

Four further covers. In each affected partial outer block $01, 30$, the inner blocks $00, 12$ were omitted entirely from F_{16}. Extend these four finite blocks with the global new row and column. They are Latin subsquares of order $m+1$: each restricted transversal supplies exactly its own old symbol block to the border, and the subsquare gains ∞. The crossing residues are $i+j+1$ in inner block 00 and $i+j$ in inner block 12, with displaced residues $2i+1$ and $2i$. Thus each has exactly the complete-graph intercalate family used in (5) and requires at least m clues of a defining subset.

Their finite block labels are

$$\begin{aligned} \text{rows: } & (0,0),(0,1),(3,0),(3,1), \\ \text{columns: } & (1,0),(1,2),(0,0),(0,2), \end{aligned}$$

where each pair records outer and inner block coordinates. Each list has four distinct members. Consequently the four enlarged subsquares intersect pairwise only at the new corner. None meets F_{16}, so they force at least $4m-3$ further clues, subtracting at most three repeated counts of that corner. An inclusion-minimal defining subset $C \subseteq U(W,T)$ is therefore critical and satisfies

$$|C| \geq |F_{16}| + 4m - 3 = 182m^2 - 18m - 44. \tag{10}$$

This proves the second bound in (1) for every odd $m \geq 3$.

Prolongation audit, checks and scope

Equivalent order form and initial values. The order-$16m+1$ result is equivalently

$$\mathrm{lcs}(n) \geq \frac{91n^2 - 326n - 5397}{128} \qquad (n \geq 49,\ n \equiv 17 \pmod{32}).$$

m	Output order $16m+1$	Protected core $\lvert F_{16}\rvert$	Final bound
3	49	1531	1540
5	81	4399	4416
7	113	8723	8748
9	145	14503	14536

Independent symbolic review. The order-$4m+1$ proof was independently reconstructed from the thesis coordinates, including the shifted transversal, every exceptional coordinate, the seven unaffected triangular blocks, and the two cover families. The order-$16m+1$ derivation received a second independent symbolic review. That review checked the $2m, 2, 2, 6$ corner-specific losses, the seven outer rectangles, the protected total, and the four enlarged subsquares. Both verdicts were **pass**; the endpoint $m = 3$ is included. A wording clarification made the binary unique-completion argument explicitly sequential: complete the two adjacent quadrants before forcing the remaining top-left quadrant. It changes no construction or count.

Supplementary finite checks. Standalone standard-library checkers implement the coordinate formulas without importing the parent construction. They verify the selected private intercalates, agreement of opposite symbols, absence of other prescribed clues, avoidance of the product transversal, and all cover-family incidences and intersections. Their saved runs report

Check	Order-$4m+1$ family	Order-$16m+1$ family
Odd parameters checked	$3, 5, \ldots, 51$	$3, 5, \ldots, 31$
Number of parameters	25	15
Private intercalates checked	162375	786385
New cover-family intercalates	24100	11420

The second reviewer audited the $16m+1$ proof symbolically and did not rerun its 15-parameter computation. These finite checks support the formulas; the uniform conclusions follow from the proofs above. The established triangular critical-set theorem remains a stated dependency.

Separate finite pruning examples. Independent integer-variable Latin-square models excluded alternative completions of four thesis-based pruned examples of sizes $94, 166, 267, 539$ at orders $13, 17, 21, 29$, respectively. All 1066 saved deletion alternatives were explicitly replayed as Latin squares agreeing with every retained clue except the designated deleted clue. These are individual critical sets, not claimed records or the basis of the uniform formulas. The order-17 example uses even $m = 4$ with a separately found transversal; it does not extend either odd-m theorem. Finite uniqueness is solver-dependent; the symbolic constructions above do not depend on that solver.

What has and has not been transferred. The earlier two sequential binary lifts give $\mathrm{lcs}(16m) \geq (377m^2 - 27m)/2$, but their chosen critical subset is not an explicit uniform mask. The proof here instead starts from the explicit seven-full-block, nine-P defining template. Its direct protected count before prolongation is $184m^2 - 9m$, smaller by $s - 7m^2 = 9m(m-1)/2$ than that sequential guarantee. No equality between these masks is assumed, and no claim that prolongation preserves criticality is needed. The theorems supply guaranteed lower bounds, not optimal values, even-m extensions, or claims of literature priority.

Reproduction. The order-$4m+1$ witness checker and audit are in `lcs_odd_prolongation/independent_audit/`; run `python3 check_thesis_shifted.py -max-m 51` there. The order-$16m+1$ files are in `lcs_other_prolongation/independent_audit/`; run `python3 check_order16m.py -max-m 31` there. Its second review is `lcs_other_prolongation/ternary_audit/order16m_second_review.txt`. The seed is Bean's thesis, Chapter 5, sections 5.3–5.4; the original prolongation section concerns intercalates rather than critical-set sizes.

Ternary products followed by prolongation

This section proves a transfer lemma and an unconditional family at orders $3 \cdot 2^k + 1$. Its input is a lower bound for *every defining subset of one specified template*. A large critical set elsewhere at the same order does not supply that input. A defining set has a unique Latin completion; an inclusion-minimal defining set is critical. Throughout, a trade means positions whose entries can be replaced by a different partial Latin square with exactly the same symbol sets in each affected row and column.

Ternary transfer lemma. Let Q be a critical set of size s in a Latin square L of order q, and let T be a transversal. Set $h = |T \setminus Q|$. Prolong L along T, and let U_T prescribe the transformed Q, all transformed transversal positions, and the entire new row and column. Suppose that every defining subset of this particular U_T contains at least b clues. Then

$$\boxed{\operatorname{lcs}(3q+1) \geq 2q^2 + 4s - 2h + b.} \tag{11}$$

Proof: a defining template. Use the outer square $C_3(i,j) = i + j \pmod 3$. Its clues $(0,0;0)$ and $(1,1;2)$ form a critical set: the missing entries are forced, successively, at

$$(0,1), (0,2), (1,0), (1,2), (2,0), (2,1), (2,2).$$

Deleting the first clue permits the global symbol interchange $0 \leftrightarrow 1$; deleting the second permits $1 \leftrightarrow 2$. Each interchange is a six-cell trade fixing the other clue.

In $M = C_3 \times L$, index rows by (i,r), columns by (j,c), and symbols by $(i + j \pmod 3, L(r,c))$. Prescribe full blocks $(0,0), (1,1)$ and a copy of Q in each of the other seven blocks; call the result P. The same seven-block order proves uniqueness. For example, block $(0,1)$ can use only outer symbol class 1, since class 0 is already exhausted in its rows and class 2 in its columns. Its Q clues then force its internal copy of L. Continue in the displayed order.

The outer diagonal has the three distinct symbols $0, 2, 1$, so its product with T is a transversal T^* of M. Prolong along T^* and prescribe the transformed P, all T^* positions, and the whole new border; call this defining template U. To prove that it defines the prolonged square, delete the known border in any completion and replace each new-symbol entry on T^* by its known displaced border symbol. The result is Latin and contains P, hence is M. This fixes the original completion too.

Three disjoint contributions. The six off-diagonal blocks are unchanged Latin subsquares and meet U in exactly Q. For each clue of Q, its criticality supplies an alternative completion after that clue is removed. Substitution inside the corresponding block changes no other clue of U. Thus all $6s$ such clues belong to every defining subset of U.

There are $q^2 - s - h$ base positions outside $Q \cup T$. Their nine outer copies form unchanged C_3 subsquares, each meeting U precisely in the two full-block positions $(0,0), (1,1)$. The outer symbol interchanges above privately protect both clues. This contributes another $2(q^2 - s - h)$ clues, all in the two full old blocks.

Finally, block $(2,2)$ together with its new row and column is an actual Latin subsquare S of order $q+1$: its symbols are outer class 1 and the new symbol. Under that relabeling, $U \cap S = U_T$. If a global defining subset $D \subseteq U$ did not define S on restriction, an alternative completion of $D \cap S$ could replace S, preserving every global row and column symbol set. This would contradict uniqueness of D. Consequently $|D \cap S| \geq b$. These cells are disjoint from both earlier contributions, including the new corner, which is counted only here. Therefore

$$|D| \geq 6s + 2(q^2 - s - h) + b = 2q^2 + 4s - 2h + b.$$

Taking an inclusion-minimal defining subset of the finite set U proves (11). The template U itself need not be critical.

The binary factor and its explicit transversal

Fix $q = 2^k$, $k \geq 2$, and a split $1 \leq a < k$. Vectors have coordinates $x_0, \ldots, x_{k-1}$ over $\mathbb{F}_2$, with x_0 the least significant bit. Write $\oplus$ for bitwise addition and & for bitwise intersection. Define the complementary masks

$$u = 2^a - 1, \qquad v = q - 2^a, \qquad M = q - 1 = u \oplus v.$$

The complete mapping used throughout this section is $y = \tau x$, where

$$\begin{aligned} y_i &= x_i + x_{i+1} && (0 \leq i < a-1), \\ y_{a-1} &= x_0 + x_{a-1} + \sum_{j=a}^{k-1} x_j, \\ y_i &= x_0 + \sum_{j=a}^{i} x_j && (a \leq i < k). \end{aligned} \tag{12}$$

All sums in the mapping formulas are in $\mathbb{F}_2$. When $a = 1$, the first line is empty and the repeated x_0 in the second line cancels.

Both τ and $p = I + \tau$ are invertible. For $y = \tau x$, put $D = \sum_{i=0}^{a-2} y_i$ and $Z = y_{a-1} + D$. Then $Z = \sum_{j=a}^{k-1} x_j$, and an explicit inverse is

$$\begin{aligned} x_0 &= y_{k-1} + Z, & x_a &= y_a + x_0, \\ x_{i+1} &= x_i + y_i \quad (0 \leq i < a-1), & x_i &= y_i + y_{i-1} \quad (a < i < k). \end{aligned}$$

For $w = px$, the inverse is

$$\begin{aligned} x_0 &= w_a, & x_{k-1} &= w_{a-1} + w_{k-1}, \\ x_i &= w_{i-1} \quad (1 \leq i < a), & x_i &= w_{i+1} + w_i \quad (a \leq i < k-1). \end{aligned}$$

These formulas also cover one-bit heads or tails, with empty index ranges.

Exactly two transversal positions are holes of the AND set. We claim

$$\{x : x \,\&\, \tau x = 0\} = \{0, u\}, \qquad \tau u = v. \tag{13}$$

For disjoint $x, \tau x$, any 1 in the first $a-1$ positions forces the next input bit to be 1, because its own output bit must vanish. If $x_0 = 0$, a first 1 in the tail would have output 1 at the same position, so the tail is zero. A first 1 in the head would propagate to $x_{a-1} = 1$, giving $y_{a-1} = 1$, again impossible. Hence $x = 0$. If $x_0 = 1$, the head is all ones. The tail has at most one 1: a second one has output 1 by the prefix-parity formula. But $y_{a-1} = 0$ requires even tail parity, so the tail is zero. This gives $x = u$; substitution yields $\tau u = v$.

The critical set in the XOR square. Let $G(r, c) = r \oplus c$, and take

$$Q = \{(r, c) : r \,\&\, c \neq 0\}, \qquad |Q| = q^2 - 3^k.$$

There are 3^k holes, from the three allowable bit pairs $(0,0), (1,0), (0,1)$. Complete holes in decreasing numerical order of $z = r \mid c$. If a candidate symbol t omits a bit of r, its true row position $(r, r \oplus t)$ is an original AND clue and excludes it; the column argument likewise forces t to contain every bit of c. Thus t contains z. If it properly contains z, its true row position is a hole of OR value $t > z$, already filled. Only $t = z$ remains. This proves uniqueness. For every clue put $d = r \,\&\, c \neq 0$. The rectangle on rows $r, r \oplus d$ and columns $c, c \oplus d$ is an intercalate whose other three positions are holes. It privately witnesses that clue's necessity. Thus Q is critical. Since τ and p are permutations, $T = \{(r, \tau r) : 0 \leq r < q\}$ is a transversal, and (13) proves $h = |T \setminus Q| = 2$ exactly.

Protected clues in the prolonged XOR template

Use $\infty = q$ for the new index and new symbol. The prolongation G^+ is given explicitly by

$$G^+(r,c) = \begin{cases} q, & c = \tau r, \\ r \oplus c, & c \neq \tau r, \end{cases} \qquad r, c < q, \tag{14}$$
$$G^+(r,\infty) = p(r), \qquad G^+(\infty, c) = \tau^{-1}(c) \oplus c, \qquad G^+(\infty,\infty) = q.$$

Each displaced transversal symbol appears once in its new row and column, so this is Latin. Let U_T contain the Q positions, all T positions, and the full new border, with their values in G^+. This is exactly the template required in the ternary lemma. It is defining: deleting the known border and restoring its known displaced symbols on T gives a completion of Q, hence the XOR square. No optimality or product conjecture is used.

For a clue $(r,c) \in Q$, its canonical private intercalate has corners

$$(r,c), \quad (r \oplus d, c), \quad (r, c \oplus d), \quad (r \oplus d, c \oplus d), \qquad d = r \,\&\, c. \tag{15}$$

If these avoid T, the intercalate survives in G^+ and meets U_T only at (r,c). Let F be the set of these protected clues. Every defining subset of U_T must contain F: otherwise swapping the corresponding intercalate gives a second completion.

Exact count of F. A transversal meets an intercalate in at most one position. Two selected corners in distinct rows and columns would be opposite corners with the same symbol. Consequently the four possible corner-hit classes in (15) are disjoint when $d \neq 0$. First count them allowing $d = 0$. Write $\mathcal{S} = \{x : x \,\&\, \tau x = 0\} = \{0, u\}$ and let wt denote bit weight.

corner hit	number	parameterization
(r,c)	q	$c = \tau r,$
$(r \oplus d, c)$	$1 + 2^{k-a}$	$x = r \oplus d \in \mathcal{S},\ c = \tau x,\ d \subseteq \tau x,$
$(r, c \oplus d)$	$1 + 2^a$	$r = x \in \mathcal{S},\ c = \tau x \oplus d,\ d \subseteq x,$
$(r \oplus d, c \oplus d)$	$q + 1$	$x \in \mathcal{S},\ d \subseteq \overline{x \mid \tau x}.$

Here $d \subseteq x$ means inclusion of bit supports, and the complement is within the k coordinates. The last three counts are respectively

$$\sum_{x \in \mathcal{S}} 2^{\mathrm{wt}(\tau x)}, \qquad \sum_{x \in \mathcal{S}} 2^{\mathrm{wt}(x)}, \qquad \sum_{x \in \mathcal{S}} 2^{k - \mathrm{wt}(x \mid \tau x)}.$$

In every class $d = 0$ occurs exactly at $(0,0)$ and (u,v). Subtracting these two cases four times, then using disjointness, gives $2q + 2^a + 2^{k-a} - 5$ unprotected AND clues. Therefore

$$|F| = q^2 - 3^k - 2q - 2^a - 2^{k-a} + 5. \tag{16}$$

How trades force additional clues. If a trade avoids F, every defining $D \subseteq U_T$ must meet its projection onto $U_T \setminus F$. Otherwise the trade's mate gives a different completion of D. Pairwise disjoint projections consequently force distinct additional clues, even if the full trades intersect at holes. We next give all the trade families used for the stronger bound. This is a bound for *every* defining subset of U_T, exactly the quantifier needed for b in (11).

Three families of pairs and the prolongation trades

The four corner parameterizations also identify all old AND clues outside F. Use the following position names:

$$\begin{aligned} D_r &= (r, r) && (r \neq 0), \\ H_d &= (u \oplus d, v) && (0 \neq d \subseteq v), \\ J_e &= (u, v \oplus e) && (0 \neq e \subseteq u), \\ T_r &= (r, \tau r) && (r \notin \{0, u\}). \end{aligned}$$

These types are disjoint and exhaust $Q \setminus F$. In addition, U_T contains T_0, T_u, so it contains every transversal position. For example, the fourth-corner hit with $x = 0$ gives exactly the nonzero diagonals D_r; with $x = u$, no nonzero d is possible. The second and third corner hits give the H and J arms; the first gives the remaining T_r.

1. Complementary diagonal pairs. For every unordered pair $\{r, s\}$ with $r \oplus s = M$, $r, s \neq 0$, except $\{u, v\}$, use rows r, s and columns r, s. In the XOR square this is

$$\begin{pmatrix} 0 & M \\ M & 0 \end{pmatrix}.$$

The off-diagonal positions are AND holes. If one lay on T, its row would belong to $\mathcal{S}$; since both coordinates are nonzero, it must be (u, v), excluded here. The diagonal positions do not lie on T, because $p = I + \tau$ has no nonzero kernel. Thus the intercalate survives, and its projection onto U_T is exactly $\{D_r, D_s\}$. There are $q/2 - 2$ pairs.

2. Complementary tail-arm pairs. For nonzero $d, e \subseteq v$ with $d \oplus e = v$, take rows $u \oplus d, u \oplus e$ and columns $0, v$. Their differences in both coordinates are v, so this is an intercalate. Its column-v positions are H_d, H_e; its column-0 positions are AND holes. No position lies on T: the unique transversal row in column 0 is 0, and that in column v is u, neither of which is used. The projection is $\{H_d, H_e\}$. There are $2^{k-a-1} - 1$ such pairs.

3. Complementary head-arm pairs. For nonzero $e, f \subseteq u$ with $e \oplus f = u$, use rows $0, u$ and columns $v \oplus e, v \oplus f$. The coordinate differences are u. The row-u entries are J_e, J_f and the row-0 entries are AND holes. The transversal columns in these rows are $0, v$, both absent. Thus this intercalate survives and projects to $\{J_e, J_f\}$. There are $2^{a-1} - 1$ such pairs.

The three families have mutually disjoint projections in the distinct D, H, J types, and together force

$$\eta = \left(\frac{q}{2} - 2\right) + \left(2^{k-a-1} - 1\right) + \left(2^{a-1} - 1\right) = \frac{q}{2} + 2^{a-1} + 2^{k-a-1} - 4 \tag{17}$$

clues outside F. The counts include zero when a head or tail has one bit; for $k = 2$ all three families are empty.

4. The q prolongation intercalates. For every $r < q$, rows r, ∞ and columns $\tau r, \infty$ contain

$$\begin{pmatrix} q & p(r) \\ p(r) & q \end{pmatrix}.$$

All four positions belong to $U_T \setminus F$. Put $Z = (\infty, \infty)$ and

$$E_r = \{T_r, (r, \infty), (\infty, \tau r), Z\}. \tag{18}$$

Distinct E_r intersect only at Z. Let V consist of all transversal and border positions. The pair projections above avoid V entirely. The next three families lie in V on projection and strengthen the requirement to hit all q sets E_r.

Cycle trades and the integer covering argument

Put $A = \tau p^{-1}$ and $B = p^{-1}$. For a permutation X fixing 0, let $c(X)$ be its number of cycles on the nonzero vectors, and set $c = c(A) + c(B) + c(\tau)$.

5. Bottom-border cycles. In rows $0, \infty$, at a nonzero column z, the entries are z and $p\tau^{-1}(z) = A^{-1}(z)$. For each nonzero orbit of A, swap these two rows' entries in the columns of that orbit. Both row symbol sets are preserved because the orbit is invariant under A^{-1}, and each column merely exchanges two symbols. This is a trade. Its row-0 positions are AND holes outside T, so its projection consists exactly of the bottom-border cells in that orbit. There are $c(A)$ disjoint projections.

6. Right-border cycles. In columns $0, \infty$, at nonzero row r, the entries are $r, p(r)$. Swapping these entries in every row of a nonzero orbit of $B = p^{-1}$ preserves column symbol sets and gives a trade. Its column-0 positions are holes outside T. The projections are exactly the nonzero right-border orbits, giving $c(B)$ disjoint sets.

7. Pure transversal cycles. For a nonzero orbit O of τ, use rows $R = \{u \oplus w : w \in O\}$ and swap symbols q, M in the two positions

$$(r, \tau r), \qquad (r, r \oplus M) \qquad (r \in R).$$

Their column sets coincide, since

$$\tau(u \oplus w) = v \oplus \tau w, \qquad (u \oplus w) \oplus M = v \oplus w.$$

The two positions are distinct and the M entry has not been displaced: otherwise $p(r) = M = p(u)$ would give $r = u$, excluded by $w \neq 0$. The M positions are AND holes, because $r \oplus M$ is the complement of r, and they are outside T. Thus the projection consists only of T_r for $r \in R$. Include every nonzero τ orbit, including the one containing u. These $c(\tau)$ projections partition the transversal positions other than T_u and include T_0.

The projections in families 5–7 are mutually disjoint: they occupy the nonzero bottom border, nonzero right border, and transversal, respectively. They avoid F, the pair projections, and Z. Every defining subset $D \subseteq U_T$ must therefore take at least c positions from V.

One more position is forced. The maps τ, A, B have no nonzero cycles of lengths one or two. Indeed

$$I + \tau = p, \qquad I + A = B, \qquad I + B = A$$

are invertible. A fixed vector lies in the kernel of $I+X$; a vector of period two lies in the kernel of $(I+X)^2 = I+X^2$, also impossible. Every nonzero orbit has at least three elements. Consequently

$$3 \leq c \leq 3 \left\lfloor \frac{q-1}{3} \right\rfloor \leq q - 1 < q. \tag{19}$$

If D used exactly c cells of V, it would use one from each cycle projection and none elsewhere in V, in particular none at Z. Each selected cell then hits at most one E_r, since the E_r intersect only at Z. It cannot hit all q of them because $c < q$. Fewer than c cells already miss a cycle. Hence $|D \cap V| \geq c + 1$.

Combining this with the protected core and disjoint projected pairs gives the required universal restriction bound

$$\begin{aligned} |D| \geq b_{k,a} &= |F| + \eta + c + 1 \\ &= q^2 - 3^k - \frac{3q}{2} - 2^{a-1} - 2^{k-a-1} + 2 + c, \qquad D \subseteq U_T \text{ defining.} \end{aligned} \tag{20}$$

This proof uses explicit trades and integer counting, without numerical optimization. Minimal pruning produces a critical set with at least this many clues, but the stronger displayed statement is what enables transfer.

The resulting family and exact-cycle refinements

Substitute $s = q^2 - 3^k$, $h = 2$, and (20) in the ternary lemma. For every $k \geq 2$ and $1 \leq a < k$, with the mapping in (12), the unconditional result is

$$\boxed{\begin{aligned}\operatorname{lcs}(3q+1) \geq 7q^2 - 5 \cdot 3^k - \frac{3q}{2} - 2^{a-1} - 2^{k-a-1} - 2 \\ + c(\tau p^{-1}) + c(p^{-1}) + c(\tau), \qquad q = 2^k.\end{aligned}} \tag{21}$$

Since each cycle count is at least one, the balanced choice $a = \lfloor k/2 \rfloor$ yields the uniform formula

$$\boxed{\begin{aligned}\operatorname{lcs}(3 \cdot 2^k + 1) \geq 7 \cdot 4^k - 5 \cdot 3^k - 3 \cdot 2^{k-1} \\ - 2^{a-1} - 2^{k-a-1} + 1, \qquad a = \lfloor k/2 \rfloor.\end{aligned}} \tag{22}$$

The powers of two penalizing the split are minimized by a balanced split. The exact-cycle expression may favor another split: every term must then be computed using that same a and that same τ. Cycle enumeration is an exact finite permutation calculation, not a fitted estimate.

q	$3q+1$	Balanced uniform	Balanced exact	Best exact over splits
4	13	60	60	60
8	25	299	299	299
16	49	1360	1366	1366
32	97	5900	5900	5903
64	193	24924	24938	24938
128	385	103550	103550	103550
256	769	425548	425614	425614

At $q = 32$, split $a = 1$ gives the value 5903. The last column maximizes only the proved expression over the finitely many splits, not over Latin squares or critical sets. The small values are not record claims; the separate $16m + 1$ construction, for example, gives the stronger 1540 at order 49. For a combined lower envelope, take the maximum of all valid constructions and verified records.

Scope of the construction. The proof prescribes a finite defining template and a lower bound that survives any pruning to an inclusion-minimal defining subset. It is an existence construction with a finite algorithm: test deletions, retaining a clue exactly when deletion destroys uniqueness. Every protected clue remains essential, and every retained clue remains essential under further deletions. There is no assertion that the initial template is critical, that this pruning is efficient at arbitrary order, or that it produces a set of exactly the displayed size.

The proof assumes neither dyadic optimality $\operatorname{lcs}(2^k) = 4^k - 3^k$ nor a conjectured product inequality. The bitwise set Q is an explicit critical set of that size, which is all that is needed. The two places where criticality matters are its private intercalates and the six unchanged product blocks. The border contribution uses a universal trade-cover bound, not optimality of the prolonged square.

In particular, a verified large critical set at order $q + 1$ cannot simply replace $b_{k,a}$ in (11). Such a record is an existence statement. The transfer requires a statement about every defining subset of the specific U_T induced in the enlarged $(2, 2)$ subsquare. This distinction is why the full protected-core and trade-cover argument has been included above.

The exact-cycle computations provide individual rigorous improvements to (22). They do not establish a closed formula for the cycle counts in all dimensions, or the sharper proposed family $\operatorname{lcs}(2^k + 1) \geq 4^k - 3^k + 2^{k+1} - 7$. The ternary transfer remains valid independently of the status of that proposal.

Which other completed products can be prolonged?

A sharp parity obstruction for the cyclic-factor $3m$ route. The product of cyclic tables $C_3 \times C_m$ has a transversal if and only if m is odd. For odd m, the diagonal gives symbols

$$(2i \bmod 3,\ 2r \bmod m), \qquad 0 \le i < 3, \quad 0 \le r < m,$$

which are distinct because multiplication by 2 is a permutation in both coordinates. Thus the diagonal is a transversal.

For even m, suppose any transversal existed and put $S = \sum_{r=0}^{m-1} r = m(m-1)/2$. In the row labels of a transversal, each second coordinate occurs three times. The same is true of its column labels and, by the transversal property, of its symbols. Summing the second-coordinate addition law gives

$$3S \equiv 3S + 3S \pmod{m}, \qquad \text{hence } 3S \equiv 0 \pmod{m}.$$

But for even m, $S \equiv m/2 \pmod{m}$, and therefore $3S \equiv m/2 \not\equiv 0 \pmod{m}$, a contradiction. This excludes *all* transversals, not only the diagonal. The notation refers to the product of the two cyclic tables; its group need not itself be cyclic when 3 divides m.

It follows that this cyclic-factor construction cannot reach an odd order $3m + 1$ by ordinary transversal prolongation: the required even m is exactly the obstructed case. There is no conflict with (22). There the even-order factor is the elementary abelian XOR square, not the cyclic table C_{2^k}, and its explicit complete mapping supplies the needed transversal for $k \ge 2$.

Two successive prolongations for odd abelian groups. Let G be an abelian group of odd order greater than one, with addition table $L(x, y) = x + y$. Multiplication by 2 is a permutation, so the diagonal (x, x) is a transversal. Prolong along it, using a new index and symbol ∞.

Choose a nonzero $a \in G$. In the prolonged square the cells

$$\{(x, x + a) : x \in G\} \ \cup\ \{(\infty, \infty)\} \tag{23}$$

form another transversal. The old cells avoid the first diagonal because $a \neq 0$, and so their entries remain $2x + a$. They meet every old row and column once and contain every old symbol once. The new corner supplies the remaining row, column and symbol. A second prolongation therefore produces a completed Latin square of order $|G| + 2$.

Taking $G = C_3^k$ gives orders $3^k + 2$, including $11, 29, 83, \ldots$ for $k \ge 2$; the same argument covers the smaller case $|G| = 3$. Taking $G = C_3 \times C_m$ for odd m gives orders $3m + 2$. These statements concern completed Latin squares and explicit transversals. They do not, by themselves, supply critical-set sizes after the second prolongation.

The reason for separating these conclusions is concrete. Prolongation changes entries on a transversal and introduces a new border. A private trade witnessing an old clue's necessity may meet that transversal and cease to be a trade of the new square. Prescribing all changed positions and the border can preserve unique completion, but a large defining set can still have a much smaller critical subset. The protected-trade and restricted-subsquare counts in the ternary proof supply the additional argument needed for a lower bound on lcs, beyond the existence of a transversal.

Finite $8m$ transversals and independent checks

The $8m$ observations are finite certificates. The absence of a transversal in the outer C_2 table does not imply its product with another Latin square lacks one. Here are explicit transversals in $C_2 \times L_m$ for $m = 1, 3, 5$, where L_m is the thesis square of order $4m$. To fix the indexing completely, write

$$L_m((R,i),(C,j)) = m(R \oplus C) + f_{R,C}(i,j), \quad 0 \le R, C < 4, \quad 0 \le i, j < m,$$

where residues are taken modulo m and

$$(f_{R,C}) = \begin{pmatrix} A & t & t & B \\ A & t & B & t \\ A & B & t & t \\ B & A & A & A \end{pmatrix}, \qquad A = i - j, \quad t = j - i, \quad B = i + j.$$

For the product, rows and columns are indexed by $a(4m) + Rm + i$, with $a \in \{0, 1\}$, and entries are $(a \oplus b)(4m) + L_m((R,i),(C,j))$. Each vector below gives the column selected in successive rows.

$$\begin{aligned} t^{(1)} &= (6,5,1,2,3,0,4,7), \\ t^{(3)} &= (14,16,8,19,4,17,15,13,5,7,11,3, \\ &\qquad 1,12,22,20,9,21,2,23,10,18,6,0), \\ t^{(5)} &= (4,5,20,3,9,31,27,15,38,36,6,19,28,7, \\ &\qquad 33,30,21,10,35,16,14,22,2,29,39,34,17, \\ &\qquad 11,37,1,32,24,25,23,18,13,8,26,12,0). \end{aligned}$$

In each case both the selected columns and the selected symbols are permutations of $0, \ldots, 8m - 1$. These are directly checkable certificates for prolongations $8 \to 9$, $24 \to 25$, and $40 \to 41$. They are not an all-m transversal formula and do not assert new critical-set sizes.

Independent audit of the ternary transfer. A separate agent reconstructed the transfer proof and found no gap under its stated universal hypothesis on b. The audit checked the outer two-clue set and its explicit deletion witnesses, the block completion order, both protected-clue classes, the exact enlarged $(2,2)$ subsquare, and the disjointness of all three counted contributions. It also checked that the already audited projected-trade-cover template is exactly U_T, so its universal restriction bound has the required scope. The coordinate proofs reproduced here establish the all-order statements; the following finite checks supplement them.

Check	Verified scope
Ternary product and private trades	$q = 4, 8, 16, 32, 64$: Latin property, product transversal, explicit completion order, every private witness in both protected classes, and entrywise agreement of the enlarged subsquare and its partial template.
Projected trade cover	All 28 dimension/split pairs for $2 \le k \le 8$: the actual trade mates, avoidance of F, exact projections, disjointness and incidence constraints.
Exact cycles and bound arithmetic	All 66 dimension/split pairs for $2 \le k \le 12$.
Other prolongations	The three $8m$ vectors above; two successive prolongations of $C_3 \times C_m$ for all twelve odd $m = 1, 3, \ldots, 23$.

The delivered files under `lcs_other_prolongation/` include `ternary_audit/independent_audit.txt`, `ternary_audit/check_ternary_transfer.py`, its JSON results, and `check_transversals.py`. The previously audited binary dependency proofs and audit comments are in `dependency_proofs/`; their explicit trade checker is `lcs_dyadic_plus_sharp_progress/check_projected_cover.py` in the delivered outputs. These checks use exact arithmetic; the ternary and transversal scripts require only the Python standard library. No claim of literature priority, optimal critical-set size, or a general unproved product-preservation principle is made.

Three simultaneous prolongations: orders $4m+3$

For every **odd** $m \geq 7$, the following new family and its safe lifts are proved:

$$\boxed{\operatorname{lcs}(4m+3) \geq S_m := \frac{21m^2 - 19m - 98}{2}.} \tag{24}$$

The same critical subset has at least $T_m = 7m^2 - 8m - 52$ clues with private intercalates. Consequently the previously audited binary lifts give

$$\operatorname{lcs}(8m+6) \geq 44m^2 - 3m - 141, \qquad \operatorname{lcs}(16m+12) \geq \frac{371m^2 + 183m - 792}{2}.$$

These follow from $q^2 + 2S_m + T_m$ and $7q^2 + 3S_m + 6T_m$, respectively, where $q = 4m+3$. All coefficients are positive: the unknown actual size of the extracted critical subset may exceed S_m. The family covers $n \equiv 7 \pmod 8$. It **does not cover 11 or 19**. In terms of the final order, (24) is $(21n^2 - 202n - 1151)/32$, for $n \equiv 7 \pmod 8$, $n \geq 31$.

Seed and three transversals. Use the order-$4m$ thesis square L and defining set P described on page 16, with masks H, J, G, F as there. For each $b \in \{0, -1, -2\}$ take the cells

row	column	symbol block and residue
$(0, i)$	$(0, m-1-i+b)$	$(0, 2i+1-b)$
$(1, i)$	$(2, i+b)$	$(3, 2i+b)$
$(2, i)$	$(3, -i+b)$	$(1, -2i+b)$
$(3, i)$	$(1, -i+b)$	$(2, 2i-b)$

with residues modulo m. Multiplication by two is invertible, so each set is a transversal. Different b give different columns in every row, so the three transversals are disjoint. They meet only blocks $00, 12, 23, 31$.

Simultaneous prolongation and contraction. For each transversal T_a, introduce one new row, column and symbol ∞_a. Replace its old entry s at (r, c) by ∞_a, and put s at (r, ∞_a) and (∞_a, c). Fill the new 3×3 corner with a Latin square on the three new symbols. The result L^+ is Latin: each old row/column exchanges three distinct old symbols for the three new ones, and recovers exactly the displaced symbols on the border; each new row/column receives every old symbol once.

Prescribe transformed P, all transversal cells and the entire border; call this defining set U. Indeed, any completion of U has all new symbols already fixed in each old row and column. Deleting the border and restoring the displaced symbols produces a Latin completion of P, necessarily L. Thus the original completion is L^+. An old private trade avoiding the three transversals survives unchanged and meets U only at its target clue. Every defining subset $C \subseteq U$ must therefore contain every clue protected by such a trade.

Counting the surviving private trades

Seven triangular blocks are untouched, and contribute $7m(m-1)/2$ protected clues. For the seven full blocks, a target $(R, i; C, j)$ has a private intercalate with other row (S, x) and other column (D, y):

(R, C)	(S, D)	residue condition
$11, 22$	00	$x = j - i + y$
33	00	$x = i - j + y$
13	02	$x = i - j + y$
23	01	$x = i - j + y$
31	20	$x = i - j + y$
32	10	$x = i - j + y$

All residues here are modulo m. For $11, 22$, take $(x, y) = (0, 0)$ when $i = j$; take $((j - i) \bmod m, 0)$ when $j = 0$ or $0 < i < j$; otherwise take $(0, (i - j) \bmod m)$. In every other block the canonical choice is $(i - j, 0)$ for $i \geq j$, and $(0, j - i)$ for $i < j$. Substitution in the square and mask shows that the opposite symbols agree and precisely the target is in P.

The only potentially affected corners are the diagonal in 00 for $11, 22, 33$; the horizontal in 12 for 13; the vertical in 12 for 32; and the target itself for $23, 31$. In $11, 22$, the forbidden 00 condition is $x + y \equiv m - 1, m - 2, m - 3$. Six targets have exactly one coordinate zero; six more have positive unequal coordinates with difference $m - 2$ or $m - 3$. Hence each loses 12. In 33, differences $m - 1, m - 2, m - 3$ contribute $2 + 4 + 6 = 12$. Each of $23, 31$ loses exactly $3m$ targets.

Flexible witnesses in block 13. Put $h = (m - 1)/2$. When $i \geq j$, the private choice is $y = 0, x = i - j$; it fails at the six targets $i \in \{0, 1, 2\}$, $0 \leq j \leq i$. When $i < j$, all admissible choices are

$$j - i \leq y \leq m - 1 - i, \qquad x = y - (j - i).$$

They are private because the two G corners have sum $i + y < m$ and the H corner has $y > x$. A choice fails only if $y - i \equiv 0, -1, -2$. The interval has $m - j$ consecutive choices, so all can fail only for $j \geq m - 3$. The complete failure list is

$j = m - 1$	$i = 0, h, h + 1$
$j = m - 2$	$i = 0, h$
$j = m - 3$	$i = h$

obtained by substituting the one-, two- and three-element intervals into the three forbidden residues. For $m \geq 7$ these are six distinct targets. Including the first six, block 13 loses exactly 12.

Completing the protected count

Flexible witnesses in block 32. The third corner is a J hole, requiring $x = 0$ or $x > y$. For $i < j$, this forces $x = 0, y = j - i$; failures have $j = m - 1, m - 2$, giving $2m - 3$ targets. For $i = j$, it forces $x = y = 0$ and fails for $j = 0, m - 1, m - 2$, giving three more. For $i > j$, every private choice is

$$0 \le y \le m - 1 - i, \qquad x = i - j + y.$$

The affected vertical corner is forbidden when $x - j \equiv 0, 1, 2$. The interval has $m - i$ consecutive choices; all fail precisely in the following four cases:

$$i = m - 1 : \ j = h, h - 1; \qquad i = m - 2 : \ j = h - 1; \qquad i = m - 3 : \ j = h - 1.$$

For example, the first row follows by solving $m - 1 - 2j \equiv 0, 1, 2$ and excluding $j = i$. The two- and three-choice intervals must start at residue 0 or 1, and at 0, respectively. Thus block 32 loses $(2m - 3) + 3 + 4 = 2m + 4$ targets.

Full block	Excluded targets	Protected targets
$11, 22, 33$ (each)	12	$m^2 - 12$
13	12	$m^2 - 12$
32	$2m + 4$	$m^2 - 2m - 4$
$23, 31$ (each)	$3m$	$m^2 - 3m$
Total	$8m + 52$	$T_m = 7m^2 - 8m - 52$

Together with the untouched triangular clues, the protected set F has

$$|F| = 7m^2 - 8m - 52 + \frac{7m(m-1)}{2} = \frac{21m^2 - 23m - 104}{2}. \tag{25}$$

All these clues belong to every defining subset of U; the T_m full-block clues retain their private intercalates in any such subset.

Canonical fallback and the small exception. Using only canonical witnesses gives losses $(3m+9)/2$ in block 13 and $(7m - 3)/2$ in block 32, hence total full-block loss $11m + 39$. The same extraction and cover argument below gives $(21m^2 - 25m - 72)/2$ for every odd $m \ge 5$. For $m = 5$, flexible block 13 loses 11 rather than 12; the protected count is 154 and the final guarantee is $154 + 13 = 167$ at order 23. These smaller values are retained as valid functions but are dominated by existing bounds at their first orders.

Why the border contributes at least $2m + 3$

The prolonged copies of old blocks 00 and 12, including their new rows and columns, are Latin subsquares S_0, S_1 of order $m + 3$. They intersect exactly in the new 3×3 corner. Their union is disjoint from F. A defining subset $C \subseteq U$ restricts to a defining set on each S_i: an alternative filling of a subsquare on the same symbol alphabet could otherwise replace it in the whole square.

Auxiliary lemma. Every defining set of a Latin square of order $t \geq 4$ has at least t clues. Fewer than $t - 1$ leave two rows empty, which can be swapped. Exactly $t - 1$ clues must occupy distinct rows, columns and symbols, since two empty labels of any type can be swapped. Relabel these clues as (i, i, i) for the $t - 1$ old labels, with one unused label ∞. A unique completion would be invariant under every simultaneous permutation of the old labels. If $t \geq 5$, the stabilizer of distinct old i, j permutes at least two other old labels. Thus the entry at (i, j) can only be i, j or ∞. The diagonal excludes i, j, forcing ∞ in every off-diagonal old cell, which repeats within a row. For $t = 4$, symmetry forces all off-diagonal old cells uniformly to use ∞, or uniformly the third old label. The first repeats ∞ in a row; the second forces ∞ in the last column of all three old rows. Both contradict Latinness. This proves the lemma.

Write $D_i = C \cap S_i$, $D = C \cap (S_0 \cup S_1)$ and $K = C \cap (S_0 \cap S_1)$. Each D_i covers all but at most one of its rows, columns and symbols. The union consequently covers at least $2m + 1$ of its $2m + 3$ rows, so $|D| \geq 2m + 1$. If $|D| = 2m + 1$, there is one clue per occupied row, so $|K| \leq 3$. But the auxiliary lemma gives $|K| = |D_0| + |D_1| - |D| \geq 2(m + 3) - (2m + 1) = 5$, a contradiction.

If $|D| = 2m + 2$, at most one clue repeats an occupied row, so $|K| \leq 4$. The lemma gives the reverse inequality. Therefore $|K| = 4$ and $|D_0| = |D_1| = m + 3$, with exactly $m - 1$ clues outside the corner in each. To cover at least $m + 2$ rows, columns and symbols, the corner must cover all three new labels of each type, and the $m - 1$ other clues must have distinct old rows, columns and symbols. This excludes both off-corner border cells and transformed transversal cells (the latter have new symbols). In S_0, since $C \subseteq U$, these remaining clues must therefore lie in the unchanged mask $1 \leq j \leq i \leq m - 1$. Distinct rows and columns force both label sets to be $\{1, \ldots, m - 1\}$; equality of their sums forces $j = i$ throughout. All entries then have old symbol zero, contradicting distinctness. Hence $|D| \geq 2m + 3$.

Choose an inclusion-minimal defining subset of finite U. It is critical, contains F, and has at least $2m + 3$ further clues in the disjoint union $S_0 \cup S_1$. Adding this to (25) proves (24). The condition $C \subseteq U$ is essential to the triangular-mask argument; no claim about arbitrary defining sets is used.

Audit, gains and the unresolved orders 11 and 19

Independent review. One agent independently reconstructed the Latin square, transversals, private rectangles and all-parameter counting argument. A second independently reconstructed the construction and checked the count and subsquare cover. The second proposed the stronger $2m + 3$ cover; the first then independently audited that new argument, including the order-four case of the auxiliary lemma and the exclusion of border and transversal clues. The parent also checked the proof and safe-lift arithmetic. Both independent coordinate programs passed for every odd $m = 3, 5, \ldots, 41$ on the applicable domains. The symbolic argument, rather than those finite checks, establishes the all-parameter theorem. No independently verified *individual* critical set of exactly S_m clues is asserted: the proof guarantees a critical subset of at least that size.

The audit files are retained under `lcs_four_m_plus_three/independent_audit/` and `lcs_four_m_plus_three/second_audit/`. In particular, `subsquare_union_2m_plus_three_audit.txt` audits the final refinement. Earlier intermediate bounds in those audit files are superseded by the final formula, with their historical scope preserved.

Order	Previous proved	New proved	Gain	Source
47	1081	1117	36	Three-transversal family
55	1485	1602	117	Three-transversal family
71	2485	2824	339	Three-transversal family
79	3081	3561	480	Three-transversal family
142	12496	12524	28	Private-intercalate binary lift
174	18792	19200	408	Lift of the order-87 protected profile
247	31578	38442	6864	Three-transversal family

Relative to the report immediately before this update, this odd-m family and its safe lifts improve the proved lower bound at 30 orders through 257. The order-87 family size is below the previous proved bound, but its guaranteed private-intercalate count still improves the lift at 174. Thus retaining profiles of nonmaximal seeds can matter.

Targeted tests at 11 and 19. Three parallel transversals in XOR squares of orders 8 and 16 were also prolonged and pruned with an exact completion oracle. The tested order-19 sets reached 195, below 245. A different explicit near-transversal repair extends Donovan's order-$2s$ square to order $2s + 1$ for odd s: a partial transversal missing two rows, columns and symbols is completed by a 3×3 Latin repair. Its Latinness and defining-set contraction were independently audited, but the tested pruned sizes reached only 55 at 11 and 186 at 19. These experiments do not improve 74 or 245 and do not prove that better choices are impossible. They are not added to the record curve.

For even $m = 2h$ with odd h, a Chinese-remainder construction produces three thesis transversals by using an outer XOR square of order eight. For $h \geq 3$ it supplies three distinct inner shifts and touches more triangular blocks. Its all-parameter protected-count pattern remains unproved; the independently replayed finite instances on the next page are included as discrete bounds. The resulting orders satisfy $4m + 3 = 8h + 3 \equiv 11 \pmod{16}$ and are at least 27. Order 11 needs distinct outer maps because $h = 1$ has only one inner shift; order 19 has $m = 4$ and requires a different even-parameter template.

Independently checked finite constructions for even m

Write $m = 2h$, with $h \geq 3$ odd. Identify an inner residue by its parity and its residue modulo h. Index the eight outer rows by $r = 2R + p$, where $R \in \{0, 1, 2, 3\}$ and p is the inner row parity. Take the permutation

$$\pi = (1, 6, 2, 5, 7, 0, 4, 3).$$

Both $\pi(r)$ and $r \oplus \pi(r)$ are permutations. If $\pi(2R + p) = 2C + p'$, choose the column parity p' and column residue modulo h to be $-i + b$ in an A or A^{T} block, and $i + b$ in a B block. The Chinese remainder theorem supplies a unique column residue modulo m. For each $b = 0, -1, -2$ this is a transversal: columns are bijective, the outer symbols form XOR of the outer labels, and the inner symbol residue has slope ± 2, invertible modulo odd h. The shifts give three distinct cells per row because $h \geq 3$.

These transversals meet blocks $00, 03, 11, 12, 20, 23, 31, 32$, leaving five triangular blocks $01, 02, 10, 21, 30$ untouched. Apply simultaneous prolongation and the same defining set U. Count the canonical private intercalates that avoid the three transversals, and add the five unchanged triangular critical sets. The following are **exact finite counts**, independently reconstructed and checked against the actual prescribed U.

m	Final order	Private intercalate clues	Protected total	Bound K_{CRT}
6	27	141	216	218
10	43	527	752	754
14	59	1137	1592	1594
18	75	1971	2736	2738
22	91	3029	4184	4186
26	107	4311	5936	5938
30	123	5817	7992	7994
34	139	7547	10352	10354
38	155	9501	13016	13018
42	171	11679	15984	15986
46	187	14081	19256	19258
50	203	16707	22832	22834

The protected set is disjoint from the new 3×3 Latin subsquare. Every defining subset of U restricts to a defining set there and must use at least two corner clues: zero or one would leave two rows empty, allowing a swap. This proves the final column by adding two to the protected count.

The first auditor constructed and checked the transversals; a second reconstructed them directly from the two congruences, checked the prolonged Latin square, and verified every counted private rectangle and the untouched triangles. The second also audited the two-clue corner addition. The independent replay is `second_audit/check_even_crt.py`, with its full counts in `second_audit/even_crt_checks.json`. The protected totals agree with $(19m^2 - 36m - 36)/2$ at the twelve listed parameters. That polynomial has **not** been proved for all m; K_{CRT} is zero outside this finite table. The bound at 91 improves 4095 to 4186, and that at 107 improves 5671 to 5938. These finite cases improve seven further orders. Together with the odd-m family and its safe lifts, this update improves 37 proved lower bounds through order 257 relative to the preceding report.

Historical conflicts with the now-refuted U

The original unrestricted product conjecture uses

$$D(n) = \max_{pq=n}\{\mathrm{lcs}(p)q^2 + \mathrm{lcs}(q)p^2 - \mathrm{lcs}(p)\,\mathrm{lcs}(q)\}.$$

In deficit form it says $\Delta(pq) \le \Delta(p)\Delta(q)$. Using only the exact $\mathrm{lcs}(4) = 7$ and verified $\mathrm{lcs}(17) \ge 200$, it would give

$$\mathrm{lcs}(68) \ge 7 \cdot 289 + 200 \cdot 16 - 7 \cdot 200 = 3823.$$

But $68^{\alpha} = 802.523777\ldots$, so

$$U(68) = 4624 - 802 = 3822.$$

Even unrestricted doubling twice gives this conflict: the hypothetical sizes at orders 17, 34 and 68 are 200, 889 and 3823, whereas the rounded uppers are 200, 889 and 3822. Rounding absorbs the first small excess over $E(17)$; multiplication amplifies it. Thus D and U cannot both be true.

The concrete 17×4 product fails minimality. The standard XOR4 product with seven full order-17 blocks and nine copies of the certified 200-clue seed has 3823 clues, but is not critical. In zero-based coordinates its clue $(18, 20; 2)$ is redundant. The corresponding seed clue $(1, 3; 2)$ has no private intercalate; the seed has 78 private-intercalate clues in total. For identical-block doubling, a full-block copy of a seed clue is essential if and only if the seed clue has a private intercalate. The group-count proof therefore establishes redundancy first at order 34 and then in the nested order-68 product. Standalone checkers replay seed uniqueness, all 200 seed deletion witnesses and the intercalate facts; direct solver queries agree. This failure concerns the specified template, not the numerical product conjecture. No 3823-clue critical set is certified. A separate independent scan of all $6^4 = 1296$ conjugate-block choices proves $\mathrm{lcs}(34) \ge 769$ by protected reduction and a safe lift of 3097 at order 68, improving the standard seed lift 3091. The stronger proved thesis bound remains 3247. Thus there is currently no verified order-68 counterexample to the rounded conjectural upper bound 3822.

Another possible counterexample. The order-33 result has settled the ceiling question negatively. A certified critical set of size at least 3823 at order 68 would give another counterexample and support the numerical product conjecture at this factorization. A defining set of that size does not suffice: every clue must also be essential. Failure of one product template does not disprove the numerical existence conjecture D.

The sharp-plus conflict is separate. The earlier proposal was

$$\mathrm{lcs}(2^m + 1) \overset{?}{\ge} 4^m - 3^m + 2^{m+1} - 7.$$

At $m = 5$ this gives 838, while $U(33) = 1089 - 255 = 834$. The size-46 and size-200 examples at orders 9 and 17 motivated that extrapolation; they did not prove it. The previous audited 838-clue tensor candidate had redundant clues. The exhaustive 6435-mask check also ruled out its specified low-mask class, not all possible order-33 constructions. Those template failures did not settle either numerical conjecture. The new 836-set first refuted U; the new 844-set also exceeds the sharp-plus value 838. Thus its lower instance at order 33 holds, while the general lower family remains open and the corresponding equality at 33 is false.

Independent audit. A separate agent checked the function domains, seed restrictions, rounding identity and both incompatibilities. It confirmed that the former joint statement was only a conjectural reformulation. A later independent audit now confirms its order-33 counterexample. The new concrete order-68 test and its independently replayable evidence are documented in `lcs_order68_product_audit/`.

Why the standard product has a redundant clue

The following group-count argument gives a solver-free test. It applies more widely to four possibly different Latin blocks. Let A, B, C, D have order m, and let Q_B, Q_C, Q_D uniquely determine B, C, D. In the Latin square

$$\begin{pmatrix} A & B+m \\ C+m & D \end{pmatrix}$$

fill A completely and use Q_B, Q_C, Q_D in the other three blocks, with the same symbol shifts. The result is uniquely completable: the full block fixes the symbol groups of its neighbours, their seeds complete them, and the remaining seed then completes its block.

Lemma. A clue in the full block is essential if and only if it has a private intercalate in this partial square. Such an intercalate uses one cell from each block.

Proof. Delete a full-block clue $(r, c; s)$. Any different completion must change it; keeping its value would complete the original defining set. All other cells in its block are fixed. The changed entry cannot be a different symbol of the same group, since the other $m-1$ entries in its block row already contain every such symbol except s. Write its new value as $m+t$.

The full-block row r now contains $m+t$ and misses s, so block B has exactly one misplaced low-group entry, equal to s, in row r, say at column j. Every other row of B remains entirely in its high group. Similarly C has exactly one misplaced s, in column c, say at row i. These two deficits force exactly one misplaced high-group entry in D, at (i, j). To identify its value, count each high symbol across B: its ordinary columns contain every high symbol once, so its exceptional column j must lack exactly $m+t$, just as its exceptional row r does. The analogous count in C shows that its exceptional row i lacks $m+t$. Hence the misplaced entry in D is $m+t$. All other entries of D have their intended low group.

Repair the three misplaced entries: replace the s in B and C by $m+t$, and replace the $m+t$ in D by s. Group counts and the Latin row/column conditions show that each repaired block is Latin on its intended symbols. The three changed positions were not seed clues, because their groups changed. Hence the repaired blocks still contain Q_B, Q_C, Q_D, and their unique completions are precisely B, C, D. The original four corner entries are thus $s, m+t, m+t, s$, and only the full-block corner is a clue. They form the required private intercalate. Conversely, switching any such private intercalate gives an alternative completion after deleting the clue. This proves the lemma.

Identical-block specialization. When all four completed blocks equal L and the three partial blocks are copies of one defining set $Q \subset L$, a full-block position outside Q has a private product intercalate. At a position belonging to Q, the lemma is equivalent to the corresponding seed clue having a private intercalate in L. Thus the standard doubling of the actual order-17 seed has $289 - 200 + 78 = 167$ essential full-block clues, and 122 redundant ones.

The concrete standard 68×68 candidate contains a copy of this redundant order-34 pattern in a partial quadrant. Its outer full block fixes the quadrant symbol groups; the remaining seed completions are unique. Deleting the indicated redundant clue therefore preserves uniqueness at order 68 as well.

Scope. The proof concerns the displayed block construction. It does not assert that a critical set of the same numerical size cannot exist in another Latin square. The accompanying package replays the seed's complete uniqueness tree and checks the intercalate obstruction without relying on an UNSAT verdict.

Conjugate-block test and a small new proved bound

The block lemma permits a cheap exact test of the 889-clue order-34 templates: all 289 full-block clues must have private cross-block intercalates. The investigation tested all $6^4 = 1296$ choices of conjugates of the given order-17 completion and its corresponding 200-clue set, with the full block placed top left. Conjugation permutes row, column and symbol coordinates. Independent row, column and symbol relabellings inside individual blocks were not searched.

A separate implementation reproduced the entire coverage histogram. The largest number of protected full-block clues was 169 out of 289, achieved by 18 choices. Thus *none of these 1296 defining sets is critical at size 889.* This is a result for the stated finite class, not for all order-34 Latin squares or all possible block constructions.

One maximizing choice is explicit. Let L, Q be the order-17 seed and let A be its row-inverse conjugate, defined by

$$A(r, L(r, c)) = c.$$

Use the full block A and the three original copies of Q in

$$P = \begin{pmatrix} A & Q+17 \\ Q+17 & Q \end{pmatrix}, \qquad \widehat{L} = \begin{pmatrix} A & L+17 \\ L+17 & L \end{pmatrix}.$$

The seed conditions prove unique completion of P to $\widehat{L}$. The checker validates 169 explicit private intercalates through full-block clues. All $3 \cdot 200 = 600$ clues in the partial blocks are essential, using the seed's deletion witnesses inside their own subsquares.

Prune redundant clues until a critical subset remains. The 769 protected clues cannot be removed, proving

$$\boxed{\mathrm{lcs}(34) \geq 600 + 169 = 769.}$$

The seed has 78 private-intercalate clues, preserved in each of the three partial blocks. Together with the 169 protected full-block clues, at least

$$3 \cdot 78 + 169 = 403$$

clues of the resulting critical set still have private intercalates. A separate enumeration of all global intercalates finds exactly 403 private clues in the initial 889-clue defining set; the reduced critical subset may gain additional private witnesses. These statements guarantee a critical set of size *at least* 769; no exact-size critical subset was extracted or asserted.

The safe doubling formula consequently gives

$$\mathrm{lcs}(68) \geq 34^2 + 2 \cdot 769 + 403 = 3097.$$

This improves the old seed lift 3091, but remains below the thesis bound 3247. Both 769 and its valid subsequent lifts are included in the numerical envelope.

Another possible ceiling counterexample. Finding a different 200-clue order-17 critical set with every clue private-intercalate essential would permit exact 2-critical doubling twice, giving 3823 at order 68. Alternatively another verified construction could reach 3823 directly. The present seed has only 78 such clues, and this investigation found no construction attaining either required product size. It therefore supplies no counterexample to U.

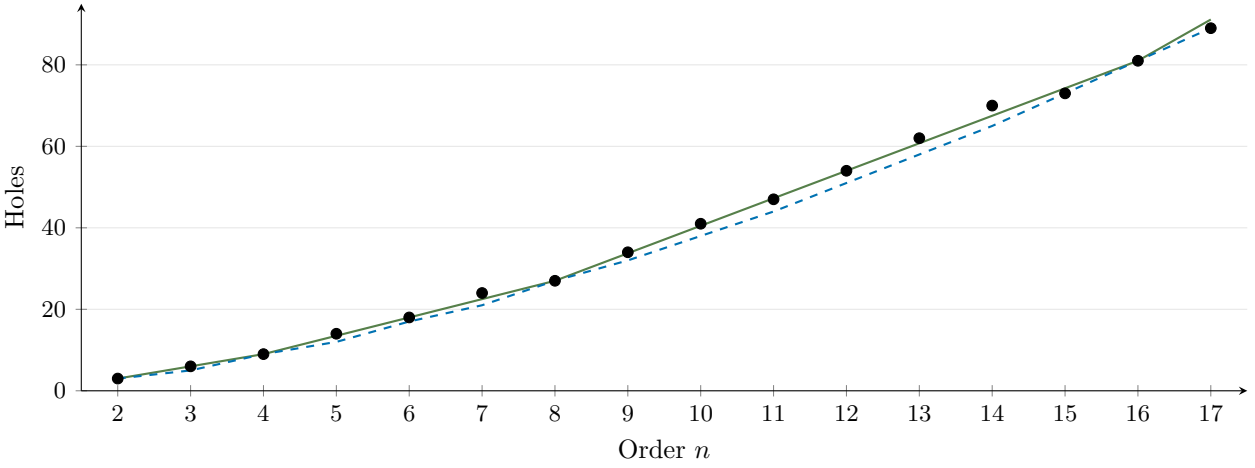
Independent verification. The package `lcs_order68_product_audit/` contains the complete 5954-node seed uniqueness tree, all 200 explicit seed deletion witnesses, the failed standard products, the conjugate coverage scan and a separately written checker. Run `check.py` and `check_conjugates.py`. The group-count lemma was independently reconstructed by the parent and auditing agents. This is a mathematical proof with finite replayable certificates, not formal verification in a proof assistant; no literature-priority claim is made.

Dyadic interpolation remains a descriptive comparison

To distinguish it from the BCC product D, write the interpolation as

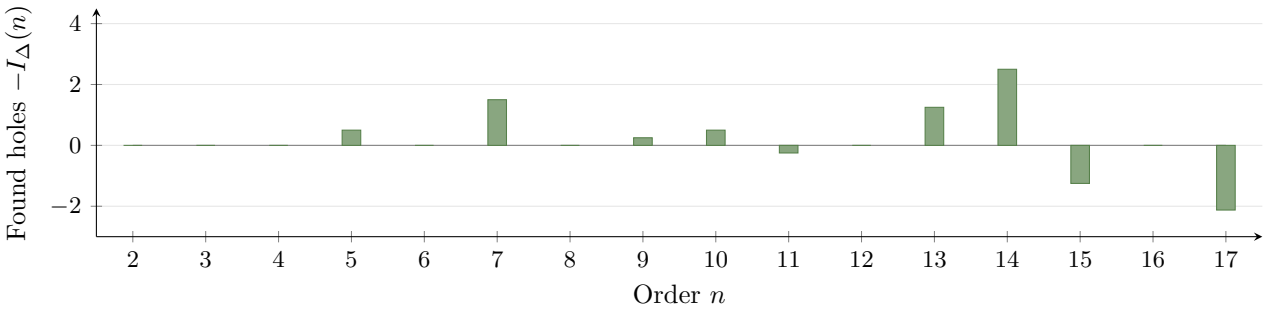
$$I_\Delta(n) = 3^m \left(\frac{2n}{2^m} - 1 \right), \qquad m = \lfloor \log_2 n \rfloor.$$

It joins the hypothetical dyadic hole counts $(2^m, 3^m)$ and is not included as a proved or conjectural bound in the revised maximum.



Solid green: I_Δ. Dashed blue: $\lfloor n^\alpha \rfloor$, the refuted minimum-hole candidate.

Black dots: exact or independently verified example holes. The order-15 size 152 has now been independently verified.



For selected records at orders 2–17, the residual is between -2.125 at order 17 and 2.5 at order 14. The verified size 47 gives residual 0.25 at order 9. Verified size 152 gives -1.25 at order 15. At order 19 the verified size 245 gives $116 - I_\Delta(19) = 4.625$. These observations concern found examples; they do not establish the unknown optima or bounded residuals at larger orders.

With $x = n/2^m$, differentiation of $(2x-1)/x^\alpha$ gives

$$n^\alpha \le I_\Delta(n) \le 1.0565189271\ldots\, n^\alpha.$$

The maximum occurs at $x = \alpha/[2(\alpha-1)]$; the upper constant is a supremum for integer n. This corrects the 1.0563 in the supplied README. That Claude-authored README is preserved as `lcs_dyadic_source_readme.txt`; its earlier fitted band and sharp-plus prediction are historical background.

Structural evidence and historical point predictions

Verified finite identifications. The earlier size-46 order-9 and size-200 order-17 completions contract to XOR tables of dimensions 3 and 4, respectively. The reproduced coordinate certificates satisfy $v(M_{ij}) = a_i \oplus b_j$ in every cell. Their transversal maps satisfy $\tau(0) = 0$ and $\tau(x \oplus y) = \tau(x) \oplus \tau(y)$; both τ and $x \mapsto x \oplus \tau(x)$ are permutations. Thus these particular completions are prolongations of elementary abelian 2-group tables along linear transversals. Their found deficits are $27 + 8$ and $81 + 8$.

A repeated frame. In the previously verified examples at orders 9, 11, 13, 17 and 22, the checker finds an empty row r, empty column c and absent symbol w, and a valid prolongation corner (p, q), such that

$$L(r, c) = L(p, q), \qquad L(r, q) = L(p, c) = w.$$

The first value is the newly introduced symbol. These are the four cells of an intercalate. The five contractions have orders 8, 10, 12, 16 and 21, respectively. The claim is existential when several valid contraction corners exist.

Evidence short of an isotopy certificate. For the earlier verified size-147 order-15 example, the row-pair cycle-type distributions in all three conjugates match those of the contracted order-16 XOR square. The supplied checker initially failed because contracted symbols were not renumbered to $0, \ldots, 14$; normalizing them in the independent checker reproduces the stated match. Matching these invariants does not prove isotopy. The README's analogous order-7 match is retained as reported evidence; that particular order-7 input was not part of this rerun.

The supplied order-17 symmetry check also reproduces a trivial stabilizer among the 180 simultaneous invertible binary-linear maps commuting with its transversal. Only the identity preserves the hole pattern in that tested class. This is not an exhaustive claim about all possible Latin-square automorphisms.

m	$2^m - 1$	Earlier =	2^m	Conditional =	$2^m + 1$	Excluded target
3	7	25	8	37	9	46
4	15	147	16	175	17	200
5	31	721	32	781	33	838
6	63	3243	64	3367	65	3488
7	127	13945	128	14197	129	14446
8	255	58467	256	58975	257	59480

The minus-point formula improves the Fu–Fu–Liao construction by one clue. Its lower inequality is proved in the separately audited package `lcs_ffl_plus_one/`, including the size-721 order-31 example. It agrees with the exact orders 3 and 7 and the earlier size-147 example at 15. The independently verified size 148 already refuted this equality at order 15; the latest size 152 strengthens the finite lower bound. In each case the construction's proved lower bound is unaffected. The former minus-point values are retained as proved lower sizes and for comparison. The plus-point formula agrees with the earlier sizes 46 and 200 at orders 9 and 17, but the new verified size 47 at order 9 refutes it as a universal equality. This does not refute that formula as a possible lower bound. The earlier independently verified size-768 order-33 example is recorded in `lcs_dyadic_plus_lower/order33_verification.json`. The latest verified 872 (page 2) exceeds 838 by 34 and the ceiling value 834 by 38. The coordinate checks do not prove the plus-point prediction for all orders. That prediction conflicted with the former upper 834, which is now refuted. The sharp-plus formula remains a research target rather than a proved term of the lower envelope.

Search diagnostic, reported rather than rerun. The supplied note reports 60-second single-core carving runs reaching size 193 in an order-17 prolongation, 163–165 in random squares, and 161 in a cyclic square. It reports an independent CP-SAT criticality check of the 193-set. The same search found only 163 and 682 in the binary group tables of orders 16 and 32, where constructions give 175 and 781. It hit a uniqueness-test node budget at order 33. These results measure that search procedure; they neither establish optimality nor test the order-33 prediction successfully. These are the original README diagnostics; the later order-19 pilot and independent verification are documented on page 42.

Earlier verified order-19 construction: 218 clues

Historical certificate. The current independently verified Bunya record is 245, with a separate certificate on page 45. This section preserves the earlier 218-clue example and its construction.

The explicit partial square $C \subset L_{19}$ on the next page has 218 clues and 143 holes. Independent computational verification gives

$$\boxed{\mathrm{lcs}(19) \geq 218.} \qquad \frac{|C|}{19^2} = 0.6038781163\ldots, \qquad \frac{19^2 - |C|}{19^{\alpha}} = 1.3444960551\ldots.$$

At discovery this improved the older reported value 205 by 13. This particular 218-set is 27 below the former candidate interval $[245, 255]$; the later verified Bunya example reaches its lower endpoint. The proved upper bound in the main review remains 298. No optimality or all-order construction is claimed.

Two explicit transversal prolongations. Let L_{17} be the completion of Richard's size-200 example, recorded in `lcs_examples.json`; it is also included in `construction.json`. All indices and symbols here are zero based. For a Latin square L_q and a permutation t such that $s_r = L_q(r, t(r))$ is also a permutation, define

$$L_{q+1}(r,c) = \begin{cases} q, & c = t(r), \\ L_q(r,c), & c \neq t(r), \end{cases} \qquad (0 \leq r, c < q),$$

$$L_{q+1}(r,q) = s_r, \qquad L_{q+1}(q, t(r)) = s_r, \qquad L_{q+1}(q,q) = q.$$

In each old row and column the displaced symbol moves to the new border and is replaced by q. The new border contains each old symbol once, so the result is Latin. Apply this first with $q = 17$, then with $q = 18$, using

$$t_{17} = (9, 14, 7, 13, 0, 15, 6, 5, 11, 8, 2, 12, 10, 1, 3, 4, 16),$$
$$t_{18} = (11, 12, 14, 3, 10, 0, 1, 13, 7, 8, 17, 16, 4, 9, 15, 6, 2, 5).$$

The independent auditor checked both transversal conditions and reversed both prolongations. The recovered 17×17 square equals the recorded L_{17} entry for entry, without relabelling. These steps construct the ambient square; the particular 218-clue subset was obtained by search and checked separately.

Discovery and independent verification. A 30-second local pilot of `lcs_hpc.c`, seed 124904, portfolio role 4, used the twice-prolonged square and the `chunk-shuffle` carving strategy. It found this set on carving 106, at approximately 25.716 seconds. The C verifier returned `CRITICAL n=19 size=218`, using 18,799,145 search nodes over all checks. An independent agent reran the Boolean exact-cover CP-SAT verifier successfully, then wrote a separate integer-cell encoding: 361 variables in $\{0, \ldots, 18\}$, with `AllDifferent` in every row and column.

For uniqueness, retain every clue and forbid the displayed complete square; the model is `INFEASIBLE`. Since that square is itself a completion, it is therefore the only one. For each clue e, remove e, retain every other clue and require the value at e to change. All 218 queries are feasible. Each returned completion was explicitly checked for Latin rows and columns, agreement with retained clues and disagreement at e. Thus every clue is essential; every proper subset omits one of these clues, proving inclusion minimality. No `UNKNOWN` status is accepted.

The separate integer encoding used OR-Tools 9.14.6206, one worker, a ten-second limit per query, and about 5.41 seconds in total. Both CP-SAT encodings share the same solver backend; the C verifier is separate. The 218 deletion witnesses are saved and directly checkable. Uniqueness rests on the exhaustive solver verdicts; no standalone UNSAT proof trace is supplied. The audit supports this finite lower bound, not a general critical-set prolongation theorem.

Exact order-19 square and critical set

Rows are labelled 0–18; the header gives columns 0–18. In the grids, `A,B,...,I` represent 10,11,...,18 and a dot is a hole. The right-hand array has exactly 218 clues; every displayed clue agrees with the left-hand square. Row 9 and column 8 are empty and symbol 1 is absent.

Complete Latin square L_{19}

```
    0 1 2 3 4 5 6 7 8 9 A B C D E F G H I
    ---------------------------------------
 0  G 1 2 3 4 5 6 7 8 H A I C D E F 0 9 B
 1  1 0 G 2 5 4 7 6 9 8 B A I C H E 3 F D
 2  2 3 0 G 6 7 4 H A B 8 9 E F I D 1 5 C
 3  3 G 1 I 7 6 5 4 B A 9 8 F H D C 2 E 0
 4  H 5 6 7 0 1 2 3 G D I F 8 9 A B C 4 E
 5  I 4 7 6 1 0 3 2 D C G E 9 8 B H F A 5
 6  6 I 4 5 2 3 H 1 E F C G A B 8 9 D 0 7
 7  7 6 5 4 3 H 1 0 F G D C B I 9 8 E 2 A
 8  8 9 A B C D E I 0 1 2 H G 5 6 7 4 3 F
 9  9 8 B A D C F E I 0 3 2 5 4 G 6 7 1 H
10  A B H 9 E F C D 2 3 0 1 6 7 4 G 5 I 8
11  B A 9 8 F E D C 3 2 1 0 H G 5 4 I 7 6
12  C D E F I 9 A B 4 5 H 7 0 1 2 3 8 6 G
13  D H F E 9 8 G A 5 I 7 6 1 0 3 2 B C 4
14  E F C H A B 8 G 6 7 4 5 2 3 0 I 9 D 1
15  F E D C H G I 8 7 6 5 4 3 2 1 0 A B 9
16  0 2 I 1 8 A B 9 C E F D 4 6 7 5 H G 3
17  4 C 8 D B I 0 5 1 9 6 3 7 E F A G H 2
18  5 7 3 0 G 2 9 F H 4 E B D A C 1 6 8 I
```

Critical set C, size 218

```
    0 1 2 3 4 5 6 7 8 9 A B C D E F G H I
    ---------------------------------------
 0  G . . 3 4 5 6 7 . H . I C D . F 0 9 B
 1  . 0 . 2 5 4 7 . . . B A I . . E 3 F D
 2  2 3 . G 6 7 4 H . B 8 . E F I D . 5 C
 3  3 G . I 7 6 5 4 . A . . . H D . 2 E .
 4  . 5 6 7 0 . . 3 . D . F 8 9 . B C 4 .
 5  I 4 7 6 . . 3 2 . C . . . . B H F . 5
 6  6 I 4 5 2 3 . . . . . G A . 8 9 D 0 7
 7  7 6 5 4 3 H . 0 . G D C B I 9 . . 2 A
 8  8 9 A . . D . I . . 2 . . 5 . 7 4 3 F
 9  . . . . . . . . . . . . . . . . . . .
10  A B H 9 E . C D . 3 . . 6 7 4 G 5 I 8
11  . A 9 . F E D C . 2 . 0 H G 5 4 I 7 6
12  C . E . I 9 A B . . . 7 0 . 2 . . 6 G
13  D . F E 9 . G A . . . . . 0 3 2 . C 4
14  E F C H . . . G . . . 5 2 3 0 I 9 D .
15  F E . . . . . . . 6 5 . . 2 . . . . .
16  . 2 I . . A B 9 . E . . 4 . 7 5 H G .
17  4 . . D B I . 5 . . . 3 7 E F A G H .
18  5 7 3 . G . 9 F . . . B . A C . 6 . I
```

Reproducible verification package. The accompanying directory `lcs_order19_verification/` contains `construction.json` (including the exact order-17 base and both vectors), `seed_n19_size218.in`, the independent audit report, `verify_order19_alldifferent.py`, and `order19_independent_certificate.json` with all 218 deletion witnesses. The standard-library checker `check_construction_and_witnesses.py` reconstructs L_{19} and validates all saved witnesses without a solver. To recompute uniqueness and essentiality, run from that directory with OR-Tools installed:

```
python3 check_construction_and_witnesses.py
python3 verify_order19_alldifferent.py \
    seed_n19_size218.in rerun_certificate.json
```

Run the independent solver checker normally, without Python's `-O` option. A rerun may choose different valid witnesses and records new timings.

Artifact identity. SHA-256 of the input file:

```
a2fe19b0fafeb813219d5373b1c48a1918ce1317b55fbfee3c15c1ca185fbd0c
```

SHA-256 of the saved independent certificate:

```
2692e2f549f77010db9f3d353e94367b218b08b6f6225ea511a2e504d6c63c39
```

The graph generator preserves this certificate in its metadata and plots the larger independently verified order-19 size 245 as a black circle. Its new certificate is separate from the 218-set above. At order 27, the verified 462 now supersedes the older reported 445 and is plotted as a black circle. The short-range interpolation plot still shows orders 2–17. Current records appear in both the campaign and full-range plots, and in the table on page 11. The record data are retained under the revised BCC curves.

Historical order-15 certificate: 148 clues

$$\boxed{\mathrm{lcs}(15) \geq 148 > 4^4 - 3^4 - 2^5 + 4 = 147.}$$

This is a counterexample to the former universal $2^m - 1$ equality at $m = 4$. It proves a lower bound, not optimality, and leaves the previously proved minus-family lower construction intact. The later size 150 raised the former candidate interval to $[150, 152]$; its certificate is on page 46, and the latest verified 152 reaches that old reference value. The order-33 lower target 838 has been surpassed by verified 872; the former upper 834 is refuted.

Search provenance. Richard supplied the full Bunya JSON: seed `297646010050`, role 2, square label `cert15-trades`, square ID 43, strategy `elite-boundary`, 9,836,486 iterations and 8,157.413 reported seconds. Direct checking reproduces 542 intercalates. The saved partial square has 148 clues and 77 holes; row 4 and column 9 are empty, and symbol 11 is absent (indices and symbols are zero based). The search label alone does not establish a general construction or isotopy.

Unique full completion L

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E
0	E	0	6	A	4	1	5	B	D	7	8	2	C	9	3
1	1	2	0	9	8	E	B	5	C	3	4	6	D	A	7
2	3	5	B	8	9	7	0	D	6	1	A	C	2	4	E
3	A	D	C	E	7	9	6	2	0	4	3	B	5	1	8
4	0	E	1	D	2	A	3	7	B	8	6	4	9	C	5
5	7	B	5	4	A	3	C	0	2	E	9	D	6	8	1
6	D	A	9	0	B	C	8	4	E	2	5	7	1	3	6
7	5	3	7	6	C	B	E	1	8	A	D	9	4	2	0
8	4	1	E	C	6	0	7	3	9	5	2	8	A	D	B
9	9	C	D	1	3	8	2	6	A	B	7	5	0	E	4
10	8	6	2	3	1	4	D	A	5	9	E	0	B	7	C
11	6	8	4	5	E	2	A	9	3	C	0	1	7	B	D
12	B	7	3	2	D	5	1	E	4	0	C	A	8	6	9
13	C	9	A	7	5	D	4	8	1	6	B	3	E	0	2
14	2	4	8	B	0	6	9	C	7	D	1	E	3	5	A

Critical partial square P

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E
0	E	.	6	A	4	.	5	.	D	.	8	2	C	9	3
1	1	.	0	9	8	.	.	5	.	.	4	6	D	A	7
2	3	5	.	8	9	7	0	D	6	.	A	C	.	4	.
3	A	D	C	E	7	9	6	2	.	.	3	.	.	1	8
4	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
5	7	.	.	4	A	3	C	0	.	.	9	D	.	.	1
6	D	A	9	0	.	C	8	.	E	.	5	7	1	3	6
7	5	3	7	6	C	.	E	1	8	.	.	9	4	2	0
8	4	1	E	C	6	0	.	3	9	.	.	8	.	D	.
9	9	C	D	.	3	8	2	6	A	.	.	.	0	E	4
10	.	6	2	3	1	.	D	.	5	.	E	0	.	7	C
11	6	8	.	.	.	2	A	9	3	.	0	1	7	.	D
12	.	7	.	2	.	.	1	E	4	.	C	A	8	6	9
13	C	.	.	7	5	D	4	8	1	.	.	3	E	0	2
14	2	4	8	.	0	6	.	C	7	.	.	E	3	5	.

Symbols $A, \ldots, E$ denote $10, \ldots, 14$; a dot denotes a hole.

Uniqueness and minimality certificate. The complete exact-cover search tree has 3,762 nodes, 190 dead leaves and exactly one completed leaf, equal to L. Each internal node selects an unfilled cell or a missing row–symbol or column–symbol constraint and branches over *every* currently possible assignment to it. The replay checker recomputes that set from the current grid. A dead leaf has no possible assignment, so exhaustive branching proves that L is the sole completion. For each of the 148 clues, an explicit alternative Latin square satisfies all other clues and changes the deleted entry. These witnesses prove every clue essential and hence establish criticality.

Independent audit. A separate agent checked the input and all 148 deletions using an integer-cell `AllDifferent` model; a Boolean exact-cover model agrees. Both use OR-Tools 9.14.6206 with one worker and finish in about 1–2 seconds. Their uniqueness queries return `INFEASIBLE` after excluding L. The separate agent also audited the exhaustive tree and witness replay. The standalone tree establishes uniqueness without relying on a solver's unsatisfiability verdict; both collections of 148 deletion witnesses are replayable using only Python's standard library.

Reproduction. The directory `lcs_order15_verification/` contains the original JSON, normalized input, complete tree, both witness collections, checkers and `independent_audit.txt`. From that directory run:

```
python3 check_certificates.py
python3 generate_uniqueness_tree.py
python3 check_certificates.py
```

SHA-256 of `order15_size148.in`:

```
f3714956dbc9dd6bef8ed09ceb28b4894551f1a6380cb170d15e7884aec10db8
```

Earlier batch: ten verified Bunya records

Directory: `lcs_bunya_verification_2026_10_15/`. The package preserves the pasted records, inputs and witnesses. Each completion is Latin, every partial entry agrees with it, the clue count is correct, and the reported intercalate count has been reproduced. These historical certificates remain valid; newer examples at orders 23 and 29 are listed on page 46. This batch established:

Order	Clues	Holes	Tree nodes	Boolean (s)	Integer (s)
18	224	100	19305	2.52	4.06
19	245	116	93774	3.71	5.34
20	281	119	67418	3.74	5.20
21	304	137	86729	5.43	7.45
22	340	144	62123	6.97	9.20
23	365	164	–	8.83	12.61
25	449	176	–	10.94	16.56
26	482	194	–	17.30	23.15
28	536	248	–	26.30	32.69
29	621	220	–	24.20	37.05

A dash means no complete standalone tree was saved; it does *not* indicate an incomplete solver check. Timings cover uniqueness and every clue deletion, with one worker per solver. They are local verification times, not search times.

What was proved computationally. For each of the ten partial squares, a Boolean exact-cover encoding excludes the supplied full completion and returns `INFEASIBLE`. A separate agent's integer-cell `AllDifferent` encoding independently returns the same result. For every single clue, both encodings produce an explicit alternative Latin square after deleting it; each alternative preserves every other clue and changes the deleted entry. All 3,847 clues therefore have necessity witnesses in each encoding: **7,694 saved and replayed witnesses**. Every solver query completed successfully; no `UNKNOWN` result was accepted. These checks establish the listed lower bounds, without asserting optimality.

Certificate strength and independent audit. The separate agent checked agreement among the pasted JSON, saved record and normalized input, reran the integer encoding, and independently replayed both witness collections. For orders 18–22 it also replayed the complete uniqueness trees: each tree branches over every legal assignment to a selected unsatisfied constraint, has only genuine dead ends, and has exactly one completed leaf, equal to the supplied square. These five cases have solver-independent uniqueness proofs. For orders 23, 25, 26, 28 and 29, uniqueness rests on the two completed OR-Tools 9.14.6206 checks. They are independent encodings using the same solver library. Separate attempts to generate standalone trees reached their 20-second cap and were not used as proof; there is no standalone UNSAT trace for these five cases. All deletion witnesses are solver-free to replay.

Construction provenance. The reported search labels describe input-file seeds at 18 and 20; a traded prolongation at 19; special seeds at 21, 25 and 29; conjugated blocks labelled 2×11, 2×13 and 2×14 at 22, 26 and 28; and a structured prolongation at 23. The exact labels, seeds, strategies, square IDs and iteration counts are preserved in the JSON as reported search metadata. The actual grids and certificates establish these finite results.

Reproduction. From the verification directory, with Python installed:

```
python3 check_certificates.py n21
python3 verify_exact_cover.py n21/input.in --output rerun_boolean.json
python3 verify_alldifferent.py n21/input.in rerun_integer.json
```

The first command uses only Python's standard library; the other two require OR-Tools. Replace `n21` by any listed order. Read the checker's output for whether uniqueness was tree-replayed or remains solver-dependent. Full grids, input hashes, `SHA256SUMS`, `verification_summary.json` and `independent_audit.txt` accompany the package. The previous order-15 and order-19 size-218 certificates are preserved separately. The latest verified bound at order 27 is 462; the older reported 445 remains historical provenance only.

Earlier batch: 15 October certificates

The four newly supplied JSON records have passed both Boolean exact-cover and integer-cell `AllDifferent` checks. They establish

$$\boxed{\text{lcs}(15) \geq 150, \quad \text{lcs}(23) \geq 366, \quad \text{lcs}(27) \geq 442, \quad \text{lcs}(29) \geq 624.}$$

These certificates remain valid and are preserved. The latest plots and tables use 367 at order 23 and 462 at order 27, the latter verified from job 30001580 (page 2), together with the separately checked $\text{lcs}(9) \geq 47$. The historical reported 445 and verified 446 are superseded by verified 462. No claim of optimality is made.

Order	Clues	Holes	Complete tree nodes
15	150	75	3224
23	366	163	173120
27	442	287	–
29	624	217	–

The order-15 tree has 130 dead leaves and the order-23 tree has 4,833; each has exactly one completed leaf, equal to the supplied square. The independent replay recomputes every branch from the current partial square. Thus these two uniqueness proofs do not depend on a solver verdict. For orders 27 and 29, both uniqueness queries finished with `INFEASIBLE` after excluding the supplied completion. Their separate 20-second attempts at exhaustive tree generation were incomplete and are *not* evidence of uniqueness. These two cases retain solver-dependent uniqueness evidence from two independently written encodings using the same OR-Tools 9.14.6206 library, with one worker per query.

For each of the $150 + 366 + 442 + 624 = 1,582$ clues, each encoding saved an alternative Latin completion after deleting that clue. All **3,164** explicit alternatives were checked against the grids: every retained clue is preserved and the deleted entry changes. Every required solver query completed; no `UNKNOWN` result was accepted. A separate agent audited the input provenance, both witness collections, the complete trees, and the structural counts. The package records the distinction between solver-independent tree replay and recorded solver verdicts.

Reported search provenance. The new 150-set came from `cert15-trades`, square 312, seed `297646010050`, role 2, `elite-light`, after 64,748,338 iterations and 58,124.228 seconds. The 366-set is labelled `prolong-structured22`, square 124; the 442-set `lab_special27-conjugate`, square 1; and the 624-set `lab_special29`, square 1. Their completed squares have respectively 546, 1,027, 648 and 3,640 intercalates, independently recounted from the supplied grids. Search labels are provenance, not proof of membership in a general construction family.

Reproduction. Directory `lcs_bunya_verification_round2/` contains the original paste, four full records, normalized inputs, certificates, independent audit and hashes. For any of the four orders run:

```
python3 check_certificates.py n15
python3 audit/check_replay.py .
python3 verify_exact_cover.py n15/input.in --output rerun_boolean.json
python3 verify_alldifferent.py n15/input.in rerun_integer.json
```

The first two commands replay saved evidence using the standard library; the last two rerun the solvers and require OR-Tools.

The order-9 update. The separate directory `lcs_order9_verification/` proves the size-47 result using a 221-node uniqueness tree (15 dead leaves, one completion) and 94 deletion witnesses. The completion has 52 intercalates, versus 64 in the old 46-set's completion, so they are not paratopic. Its unique ordinary contraction is an order-8 square with 19 intercalates, rather than the XOR table's 112. Consequently the historical XOR-prolongation observations for the old 46-set must not be attributed to the new 47-set.

Verified milestone: 151 at order 15

Result and provenance. The follow-up Bunya campaign gives

$$\boxed{\mathrm{lcs}(15) \geq 151.}$$

This improves the previous independently verified 150-set by one. The supplied source is array 29909858, task 20, rank 1, record line 3. The growth worker reports iteration 89,139 and about 84,281 seconds. Its inherited seed metadata is `cert15-trades`, square 312, `elite-light`; the inherited 64,748,338 iterations describe the earlier search state, not the growth worker's iteration count.

The actual change. The full completion is exactly the same as for the 150-set, with 546 intercalates. In zero-based row–column–symbol coordinates, remove $(7, 13, 4)$ and add $(10, 3, 6)$ and $(11, 9, 7)$. Thus the new partial has 151 clues and 74 holes. Row 8 and column 4 are empty; symbol 12 is absent. This is a finite critical-set construction, not a claimed all-order formula or an optimality proof.

Independent verification. A fresh Boolean exact-cover encoding and a separate integer-cell AllDifferent encoding each exclude every completion except the supplied one. More strongly, a complete Algorithm X tree has 2,935 nodes, 123 dead leaves and exactly one completed leaf. A separate standard-library checker recomputes every branch from the partial square and checks that the completed leaf equals the supplied grid. Uniqueness therefore has a replayable exhaustive certificate, independent of trusting either solver verdict. Both encodings also produce an alternative Latin completion for every single-clue deletion. All 302 new alternatives, and all 151 supplied alternatives, preserve every other clue and change the deleted entry: 453 witness checks in total. This update uses independent verification programs and certificate replay; it makes no claim of a new separate-agent proof audit. Earlier proof audits remain unchanged.

Protected lift profile. Direct enumeration gives the coherent same-seed tuple $(m, s, t_2, t_3) = (15, 151, 54, 3)$. Consequently the audited H_2, H_3, H_4 formulas give 581 at order 30, 1211 at order 45 and 2352 at order 60. These do not improve the respective existing bounds. In particular, the old 150-set has $t_3 = 6$ and gives 1212 at order 45; its witnesses cannot be combined with size 151. All old profiles are retained.

Reproduction. The full source, both solver certificates, all witnesses and the exhaustive tree are in `lcs15_151_bunya_29909858/`. From that directory, `python3 check_certificates.py n15` replays the tree and both freshly generated witness collections without a solver. The separate `verify_supplied_witnesses.py` checks the supplied collection and local lift profile.

The order-15 size-151 example

Unique full completion L

```
E C D 5 B 4 0 1 6 2 A 3 7 9 8
7 8 A 2 9 1 6 3 0 D 5 4 E B C
C E 2 A 4 B 3 6 1 5 D 9 8 0 7
6 1 8 B C A 7 5 E 9 4 D 0 2 3
2 D C 7 0 3 B 4 9 E 8 1 5 6 A
A 5 7 C 1 6 9 B 4 8 E 0 D 3 2
9 A 1 0 7 8 5 D 2 6 3 C B E 4
8 7 5 D 6 9 1 0 3 A 2 B C 4 E
4 B 0 1 5 E D C A 3 6 7 2 8 9
D 2 E 8 3 0 4 9 B C 7 6 A 1 5
B 4 3 6 E C 2 A 5 0 1 8 9 7 D
1 6 9 4 A 5 8 E C 7 B 2 3 D 0
5 9 6 3 8 7 A 2 D 1 0 E 4 C B
3 0 B E D 2 C 7 8 4 9 5 1 A 6
0 3 4 9 2 D E 8 7 B C A 6 5 1
```

Critical partial square P

```
E . D 5 . 4 . . . 2 A 3 7 9 8
7 8 A 2 . . 6 3 0 D . 4 E B .
. . 2 A . B 3 6 1 . D 9 8 0 7
6 1 . B . A 7 5 E 9 4 D . 2 .
2 D . 7 . 3 B . 9 E 8 . 5 6 .
. 5 7 . . 6 9 B 4 8 E 0 D 3 2
9 . 1 0 . 8 5 . . 6 3 . B E 4
8 7 5 D . 9 . 0 3 . . B . . E
. . . . . . . . . . . . . . .
D 2 E 8 . . 4 9 B . 7 . A 1 5
B 4 3 6 . . 2 A 5 0 1 8 9 . D
1 6 . 4 . . 8 E . 7 B 2 3 D 0
5 9 6 . . 7 A 2 D 1 0 E 4 . .
3 0 B . . 2 . 7 . 4 9 5 1 A 6
0 . 4 9 . D . 8 7 B . A 6 5 1
```

Symbols A–E represent 10–14; dot denotes a hole. All indices start at 0. The full square and partial are read directly from the supplied JSON. There are 151 clues and 74 holes; the completion has 546 intercalates.

Relative to the preceding 150-set on this same completion, delete $(7, 13, 4)$ and add $(10, 3, 6)$ and $(11, 9, 7)$. The old and new partials therefore differ at three cells; neither partial contains the other.

Input fingerprint. SHA256 of `lcs15_151_bunya_29909858/n15/input.in`:

```
e43d7c418897840a17f7cc60db6ebf974de3b3de4ad230f6fd2f34a6d9617e88
```

The earlier 150-set and its certificate are preserved on the following page as historical evidence. The record table and all current graphs use 151.

Historical order-15 size-150 example

Unique full completion L

```
   0 1 2 3 4 5 6 7 8 9 A B C D E
   -----------------------------
 0 E C D 5 B 4 0 1 6 2 A 3 7 9 8
 1 7 8 A 2 9 1 6 3 0 D 5 4 E B C
 2 C E 2 A 4 B 3 6 1 5 D 9 8 0 7
 3 6 1 8 B C A 7 5 E 9 4 D 0 2 3
 4 2 D C 7 0 3 B 4 9 E 8 1 5 6 A
 5 A 5 7 C 1 6 9 B 4 8 E 0 D 3 2
 6 9 A 1 0 7 8 5 D 2 6 3 C B E 4
 7 8 7 5 D 6 9 1 0 3 A 2 B C 4 E
 8 4 B 0 1 5 E D C A 3 6 7 2 8 9
 9 D 2 E 8 3 0 4 9 B C 7 6 A 1 5
10 B 4 3 6 E C 2 A 5 0 1 8 9 7 D
11 1 6 9 4 A 5 8 E C 7 B 2 3 D 0
12 5 9 6 3 8 7 A 2 D 1 0 E 4 C B
13 3 0 B E D 2 C 7 8 4 9 5 1 A 6
14 0 3 4 9 2 D E 8 7 B C A 6 5 1
```

Critical partial square P

```
   0 1 2 3 4 5 6 7 8 9 A B C D E
   -----------------------------
 0 E . D 5 . 4 . . . 2 A 3 7 9 8
 1 7 8 A 2 . . 6 3 0 D . 4 E B .
 2 . . 2 A . B 3 6 1 . D 9 8 0 7
 3 6 1 . B . A 7 5 E 9 4 D . 2 .
 4 2 D . 7 . 3 B . 9 E 8 . 5 6 .
 5 . 5 7 . . 6 9 B 4 8 E 0 D 3 2
 6 9 . 1 0 . 8 5 . . 6 3 . B E 4
 7 8 7 5 D . 9 . 0 3 . . B . 4 E
 8 . . . . . . . . . . . . . . .
 9 D 2 E 8 . . 4 9 B . 7 . A 1 5
10 B 4 3 . . . 2 A 5 0 1 8 9 . D
11 1 6 . 4 . . 8 E . . B 2 3 D 0
12 5 9 6 . . 7 A 2 D 1 0 E 4 . .
13 3 0 B . . 2 . 7 . 4 9 5 1 A 6
14 0 . 4 9 . D . 8 7 B . A 6 5 1
```

Symbols $A, \ldots, E$ mean $10, \ldots, 14$; dots are holes. All indices are zero based. Row 8 and column 4 are empty, and symbol 12 is absent. There are 150 clues and 75 holes. The full grid has 546 intercalates, compared with 542 for the previous 148-set's completion; the completions therefore cannot be isotopic or paratopic.

What this suggests for generalization. A larger seed need not give a better protected product lift. Let s be the size, t_2 the number of clues with a private intercalate witness, and t_3 the number with a private order-three witness as defined in the audited lift construction. Direct local enumeration, independently repeated, gives:

Order-15 seed	s	t_2	t_3	H_2 at 30	H_3 at 45	H_4 at 60
Earlier 147-set	147	132	15	651	1215	2808
Previous 148-set	148	58	4	579	1198	2367
New 150-set	150	54	6	579	1212	2349

Here $H_2 = 225 + 2s + t_2$, $H_3 = 450 + 5s + 2t_3$, and $H_4 = 1575 + 3s + 6t_2$. Each row uses the witnesses of that same seed; the size 150 cannot be combined with the earlier 147-set's $t_2 = 132$. Thus the new record improves the finite bound at 15, but these tested lifts do not improve the bounds at 30, 45 or 60. This is a limitation of the specific lifting methods, not evidence against other generalizations.

New consequences from the other seeds. The 366-set at order 23 has $t_2 = 28$, $t_3 = 0$; the 624-set at order 29 has $t_2 = 67$, $t_3 = 0$. Applying the existing audited lift formulas gives

$$\begin{aligned} &\mathrm{lcs}(46) \geq 1289, \qquad \mathrm{lcs}(58) \geq 2156, \quad \mathrm{lcs}(69) \geq 2888, \\ &\mathrm{lcs}(87) \geq 4802, \qquad \mathrm{lcs}(207) \geq 24614. \end{aligned}$$

The first two use H_2; orders 69 and 87 use H_3. Two ternary lifts from order 23 give $32 \cdot 23^2 + 21 \cdot 366 = 24614$. These improve the previous report's construction bounds, respectively 1288, 2083, 2883, 4787 and 23937, and enter its proved envelope. The later 367-set improves the order-69 and order-207 consequences to 2893 and 24635 (page 50); the 366-set still supplies 1289 at order 46. No claim about worldwide record priority is made.

The then-proposed interval at order 15 was $[150, 152]$; its upper endpoint is now retained only as a reference from the refuted global conjecture. The old minus-point equality predicting 147 is false, while its audited lower construction remains valid. Likewise 47 at order 9 refutes the old plus-point equality predicting 46. Neither finite improvement establishes a replacement all-order equality.

Certificate identity. SHA-256 of `lcs_bunya_verification_round2/n15/input.in`:

`49958348cbc8ae3cd8052d804703db14df2363772857cc0d09826e58088a35ae` The full matrices in this page are generated directly from that record; the earlier size-148 matrices and certificates are retained unchanged in the historical appendix and their original verification package.

Earlier certificates: 16 October 2026

These historical Bunya records establish

$$\boxed{\mathrm{lcs}(23) \geq 367, \qquad \mathrm{lcs}(27) \geq 446.}$$

They improve the previous independently verified sizes 366 and 442. The size 446 also exceeds the older reported 445, so order 27 now appears as a verified black circle in every current plot. The original 445-set has not been independently checked; its provenance is retained separately. These are finite lower bounds, without an optimality claim.

Order	Clues	Holes	Complete tree nodes
23	367	162	197670
27	446	283	–

Uniqueness and minimality. For both partial squares, a Boolean exact-cover model and a separately written integer-cell `AllDifferent` model exclude the supplied completion and return `INFEASIBLE`. Both use OR-Tools 9.14.6206 with one worker per query. Every one of the $367 + 446 = 813$ clues also has an explicit alternative completion after deletion in each encoding: **1,626 saved witnesses**. Each alternative is Latin, preserves every other clue and changes the removed entry. All required solver queries completed successfully; no `UNKNOWN` result was accepted.

The order-23 uniqueness tree has 197,670 nodes, 7,087 dead leaves and exactly one complete leaf. A separate checker recomputes all legal branches from the partial grid, proving uniqueness without relying on a solver's unsatisfiability verdict. At order 27, the standalone tree-generation attempt reached its 20-second cap without completing; it is not used as evidence. Uniqueness for this case rests on the two completed solver checks, which use different encodings but share the same solver library. All deletion witnesses for both records are solver-free to replay.

Independent audit and provenance. A separate agent checked the original paste against the saved JSON and text inputs, independently replayed all 1,626 witnesses and the complete order-23 tree, reviewed the solver encodings, and counted the local trade witnesses. The 367-set has search label `prolong-structured22`, square ID 178, seed `297646010117`, role 5, strategy `chunk-shuffle`, 68,118 iterations and 84,799.216 reported seconds. Its completion has 959 intercalates, compared with 1,027 for the previous 366-set's completion. The 446-set has label `lab_special27-conjugate`, square ID 1, seed `297646010145`, role 1, strategy `chunk-shuffle`, 6,873 iterations and 85,299.222 reported seconds. It uses the same 648-intercalate completion as the earlier 442-set.

Consequences for construction bounds. The actual profiles (m, s, t_2, t_3) of the new seeds are $(23, 367, 25, 0)$ and $(27, 446, 116, 21)$. The earlier profiles $(23, 366, 28, 0)$ and $(27, 442, 129, 27)$ remain available; a larger critical set need not have more private witnesses. The existing ternary lift and its second iterate give

$$\mathrm{lcs}(69) \geq 2 \cdot 23^2 + 5 \cdot 367 = 2893, \qquad \mathrm{lcs}(207) \geq 32 \cdot 23^2 + 21 \cdot 367 = 24635.$$

These improve the report's previous construction bounds 2888 and 24614. The old 366-set still gives the better binary lift at order 46: $529 + 2 \cdot 366 + 28 = 1289$, whereas the new seed gives 1288. The independent audit confirmed the complete numerical envelope changes through order 257. No witness counts from different seeds are combined.

Reproduction. Package `lcs_bunya_verification_round3/` contains the original paste, grids, certificates, audit and checksums. From that directory run:

```
python3 check_certificates.py n23
python3 check_certificates.py n27
python3 audit/check_replay.py .
```

These commands replay saved evidence using Python's standard library. The included solver scripts can separately regenerate the two encodings.

Earlier order-33 record 831 and protected lifts

This section records the earlier 831 milestone, followed by 836 on page 52, then 841 on page 54, then 844 on page 55, and now 872 on page 2. The completed 24-hour Bunya pilot retained a search-certified maximum of 768, but its pending output contained a uniquely completable 875-clue set on the stored `lab_special33` square (array 29814974, task 2, rank 1). All 32 saved pending sets were noncritical. Sequentially removing proved redundant clues, with randomized deletion-order seed 20261021, removed 44 clues from the 875-set and gave

$$\boxed{\mathrm{lcs}(33) \geq 831.}$$

Eight randomized repairs of this same input gave certified sizes $818, 819, 819, 823, 823, 824, 830, 831$; an initial index-order repair gave 829. These finite searches do not establish a maximum for this square or input.

Independent verification. A Boolean exact-cover model excluded every different completion in 5.24 seconds. A separately written integer `AllDifferent` model independently returned infeasible in 6.34 seconds. The independent audit also checked the completion, source hash, exact 44-clue deletion log, and all 831 alternate-completion witnesses, one for each clue. The latter are solver-free to replay. Uniqueness still relies on completed solver checks; no portable exhaustive uniqueness tree is claimed.

Why the pending size was not a record. A limited deletion check leaves the clue present when its status is unknown. Hence pending sets can contain redundant clues. Once a partial P has unique completion L, an alternative completion after removing clue x must change x: otherwise it also completes P, contradicting uniqueness. An independently audited C experiment exploits this restriction. Under the same 2-second/10-million-node cap it resolved 20 of 32 selected known-redundant tests; the original oracle resolved none. This is a selected diagnostic, not an overall speedup estimate. The original C verifier also returned unknown on the valid 831 input at its node limit.

Keep separate construction profiles. The exact audited seeds give:

Seed size s	Private-intercalate clues t_2	$H_2(33)$ at 66	$H_4(33)$ at 132
768	758	3383	14475
830	420	3169	12633
831	403	3154	12534

Here $H_2 = m^2 + 2s + t_2$ and $H_4 = 7m^2 + 3s + 6t_2$ are the previously audited protected lifts. Thus the old 768-seed, with all 768 essentiality witnesses now renewed, improves the report's earlier construction values 2673 and 12375 to 3383 and 14475. A larger critical set need not be the stronger product seed. No profile combines the size of one set with another's witnesses. The new profiles use $t_3 \geq 0$ only; no exact ternary-witness count is claimed. All resulting mixed and recursive lifts are included in the updated data. An independent recomputation through order 257 confirms the only changed construction entries above 33: $66 \mapsto 3383$, $99 \mapsto 6333$, $132 \mapsto 14475$ and $198 \mapsto 27143$ at that milestone. The 836 seed subsequently raised the order-99 entry to 6358, the 841 seed to 6383, the 844 seed to 6398, and the latest 872 seed to 6538. The audit is saved in `lcs33_review_29814974/independent_audit/envelope_update.json`.

At the 831 milestone, the ceiling 834 was still compatible with the record. The subsequent 836 example refutes it. The current unconditional bounds are $872 \leq \mathrm{lcs}(33) \leq 977$. The table here preserves the three earlier profiles; the subsequently verified 836 profile is $s = 836, t_2 = 396$.

Files. `lcs33_review_29814974/n33_831.in` contains the square and partial; `n33_831.json` also contains every deletion witness. The package includes both solver checks, separate agent audit and source archive. From its directory run `python3 replay_record.py n33_831.json` to replay essentiality; add `-check-unique -seconds 60` to rerun the solver-dependent uniqueness check. OR-Tools is needed only for the latter.

First ceiling counterexample: 836 at order 33

This historical certificate is retained; the latest verified size is 872 (page 2). The standalone certificate is `lcs33_review_29814974/n33_836.json`; its input grid is `n33_836.in`. The source is the same archived 875-clue pending set, `task_2_n33/attempt_0/rank_1.pending.in`, on the `lab_special33` completion. That completion has 6976 intercalates. The input hash and exact 39 deleted cell indices are checked in the independent audit and retained with the record.

How the new repair differs. Before the randomized scan, protect any clue having a private intercalate and retain its four-cell alternate completion. With Python RNG seed 20261021, shuffle the remaining clues. For a proposed deletion force the deleted cell to change and ask the Boolean exact-cover CP-SAT model for a completion, with a one-second solver limit. Delete only on infeasibility; otherwise keep a validated alternate completion or mark unknown. A second pass permits four seconds per unresolved query. This bounded local trial completed in 53.18 seconds: 586 deletion queries, 39 deletions, no unknown outcomes, and 836 essentiality witnesses. The worker invariant and its implementation were separately audited.

Criticality evidence. The final partial admits its stored Latin completion. Fresh uniqueness tests returned infeasible in 2.43 seconds (Boolean exact cover) and 2.75 seconds (integer AllDifferent). A separately written independent agent checker confirmed uniqueness in 2.98 seconds and reconstructed all 836 alternate Latin squares, each agreeing with every remaining clue except the one removed. It also checked Latin validity, exact size, source membership and deletion provenance. The witnesses have trade sizes from 4 to 131. All 396 private-intercalate clues are counted on this same partial; there are 508 private intercalates in total.

The witness replay proves that every clue is necessary. Together with the completed uniqueness checks, this certifies a critical set of size 836. Uniqueness remains solver-dependent: no portable UNSAT proof is claimed. The independent checker and frozen input are in `lcs33_review_29814974/independent_audit/order33_size836/`. Run `python3 verify.py -seconds 120` there with OR-Tools installed; the fixed certificate does not depend on rerunning the heuristic search. Solver-free replay is available from the review directory as `python3 replay_record.py n33_836.json`.

Consequence. To 80-digit working precision,

$$33^2 - 33^{\log_2 3} = 833.854653663322447366\ldots, \qquad \mathrm{lcs}(33) \geq 836 > 834 = \lceil E(33) \rceil.$$

This disproves the proposed universal ceiling upper bound by two clues. It does not determine lcs(33), disprove dyadic exactness, prove the unrestricted product formula, or establish the sharp-plus family. At this milestone the target 838 was two above the record. The subsequent 841-set now exceeds it by three.

Historical proved lifts. The 836-set has coherent profile $(s, t_2) = (836, 396)$. It gives $H_2(33) = 3157$, $H_4(33) = 12507$, both weaker than the old 768 seed's 3383 and 14475. With the safe lower count $t_3 \geq 0$, however,

$$\mathrm{lcs}(99) \geq 2 \cdot 33^2 + 5 \cdot 836 = 6358,$$

which improves the 831 seed's 6333 by 25. The lower-envelope data use each profile separately and keep the better construction at each order.

Earlier exact order-33 example: 844

Symbols are $0, 1, \ldots, 9, A, \ldots, W$ (values 0–32). A dot is a hole. The right-hand partial has exactly 844 clues and 245 holes. All its entries agree with the left-hand completion. Machine-readable grids and all 844 witnesses are retained in the certificate, avoiding transcription reliance.

Unique full completion L

```
   0 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q R S T U V W
   -----------------------------------------------------------------
 0 W 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q R S T U V 0
 1 1 0 W 2 5 4 7 6 9 8 B A D C F E H G J I L K N M P O R Q T S V U 3
 2 2 3 0 1 W 7 4 5 A B 8 9 E F C D I J G H M N K L Q R O P U V S T 6
 3 3 2 1 0 7 6 W 4 B A 9 8 F E D C J I H G N M L K R Q P O V U T S 5
 4 4 5 6 7 0 1 2 3 W D E F 8 9 A B K L M N G H I J S T U V O P Q R C
 5 5 4 7 6 1 0 3 2 D C W E 9 8 B A L K N M H G J I T S V U P O R Q F
 6 6 7 4 5 2 3 0 1 E F C D W B 8 9 M N K L I J G H U V S T Q R O P A
 7 7 6 5 4 3 2 1 0 F E D C B A W 8 N M L K J I H G V U T S R Q P O 9
 8 8 9 A B C D E F 0 1 2 3 4 5 6 7 W P Q R S T U V G H I J K L M N O
 9 9 8 B A D C F E 1 0 3 2 5 4 7 6 P O W Q T S V U H G J I L K N M R
10 A B 8 9 E F C D 2 3 0 1 6 7 4 5 Q R O P W V S T I J G H M N K L U
11 B A 9 8 F E D C 3 2 1 0 7 6 5 4 R Q P O V U W S J I H G N M L K T
12 C D E F 8 9 A B 4 5 6 7 0 1 2 3 S T U V O P Q R W L M N G H I J K
13 D C F E 9 8 B A 5 4 7 6 1 0 3 2 T S V U P O R Q L K W M H G J I N
14 E F C D A B 8 9 6 7 4 5 2 3 0 1 U V S T Q R O P M N K L W J G H I
15 F E D C B A 9 8 7 6 5 4 3 2 1 0 V U T S R Q P O N M L K J I W G H
16 G H I J K W M N O P Q R S T U V 0 1 2 3 4 5 6 7 8 9 A B C D E F L
17 H G J I L K N W P O R Q T S V U 1 0 3 2 5 4 7 6 9 8 B A D C F E M
18 I W G H M N K L Q R O P U V S T 2 3 0 1 6 7 4 5 A B 8 9 E F C D J
19 J I H W N M L K R Q P O V U T S 3 2 1 0 7 6 5 4 B A 9 8 F E D C G
20 K L M N G H I J S T U V O W Q R 4 5 6 7 0 1 2 3 C D E F 8 9 A B P
21 L K N M H G J I T S V U P O R W 5 4 7 6 1 0 3 2 D C F E 9 8 B A Q
22 M N K L I J G H U W S T Q R O P 6 7 4 5 2 3 0 1 E F C D A B 8 9 V
23 N M L K J I H G V U T W R Q P O 7 6 5 4 3 2 1 0 F E D C B A 9 8 S
24 O P Q R S T U V G H I J K L M N 8 9 A B C W E F 0 1 2 3 4 5 6 7 D
25 P O R Q T S V U H G J I L K N M 9 8 B A D C F W 1 0 3 2 5 4 7 6 E
26 Q R O P U V S T I J G H M N K L A W 8 9 E F C D 2 3 0 1 6 7 4 5 B
27 R Q P O V U T S J I H G N M L K B A 9 W F E D C 3 2 1 0 7 6 5 4 8
28 S T U V O P Q R K L M N G H I J C D E F 8 9 A B 4 5 6 7 0 W 2 3 1
29 T S V U P O R Q L K N M H G J I D C F E 9 8 B A 5 4 7 6 1 0 3 W 2
30 U V S T Q R O P M N K L I J G H E F C D A B 8 9 6 W 4 5 2 3 0 1 7
31 V U T S R Q P O N M L K J I H G F E D C B A 9 8 7 6 5 W 3 2 1 0 4
32 0 J 3 G 6 L 5 M C V F S A P 9 Q 0 B R 8 U D T E K 7 N 4 I 1 H 2 W
```

Critical set P, size 844

```
   0 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q R S T U V W
   -----------------------------------------------------------------
 0 . . . 3 . 5 6 . . . . . C D . F . . . . K L M N . . . . S T U V 0
 1 . . W . 5 4 7 6 . 8 B A . C . E . G J I L K N M . 0 . Q T . V U 3
 2 2 3 0 1 . . 4 5 A B . . . F . D I . G H M N . . . R O . U V S T .
 3 3 2 1 0 . 6 W 4 B A . 8 F E . C J I H G N M L K R Q . . . U T S 5
 4 4 5 6 7 0 1 2 3 W D E . 8 . . B . L M N G H I J S T U V O P Q R .
 5 . 4 7 6 . . 3 2 D C . E . 8 . A L K . M H G J I . . V U P . R Q F
 6 6 7 4 5 2 3 0 1 E . C D . . . . M N K L I . G H U V S T . R O P A
 7 . . . . . .
 8 8 . A B C D E F 0 1 2 3 4 5 . 7 W P Q R S T U V G H I J K L M . 0
 9 . . B A D C F E 1 0 3 2 5 4 . 6 P O W Q . . V U . G J I L K N M R
10 A B 8 . E F C D 2 . 0 1 6 7 . 5 Q R O . W V S T I J G H M N K L U
11 B A . 8 . E D C 3 2 . 0 7 . . . R Q P O V . W S J I H G N M L . T
12 C D E F 8 . A B . 5 . 7 0 1 . 3 S . U V . P Q R W L M N G H I J K
13 . C F E . 8 B A 5 4 7 6 1 0 . 2 T S V . P O R Q L K . M H . J I N
14 E F . . A B 8 . 6 7 4 5 2 3 . 1 U V S T . R O P M N K . W J G . I
15 F E D . B A . . 7 . 5 4 3 2 . 0 V U T . . Q P O N M L K J I . G H
16 G H I J K W M N O P Q R S T . V 0 1 2 3 4 5 6 7 8 . . B C D E F .
17 H G . I L . N W P O R Q T . . U 1 0 3 2 5 . 7 6 . 8 B A D C F E M
18 I . G H M N K L . R O P U V . T . 3 0 1 6 7 4 5 A B 8 . E F C D J
19 J I H W N M L K R Q P . V U . S 3 2 1 0 7 6 5 4 B A . 8 F E . C G
20 K L M N G H I J S . U V O W . R 4 5 6 7 0 1 2 3 C D E F 8 . . B P
21 L K N M H G J I T S V . P O . . 5 4 7 6 1 0 . 2 D C F E . 8 B A Q
22 M N K L I J G H U W S T Q . . P 6 7 4 5 2 3 0 1 E F . D A B 8 . V
23 N M L . . I H G V U T W . Q . O 7 6 5 4 3 2 1 0 F E D C B A . 8 .
24 O P Q R S T U V G H I J K L . . 8 . A B C W E F 0 1 2 3 4 . 6 7 .
25 P O . . T S V U H G . . L K . M . 8 B A D C . W 1 0 . 2 5 4 7 6 E
26 Q R O P U V S T I J G H M N . L A W 8 . E F C . 2 3 0 1 6 7 4 5 B
27 R . P O V U T S J I H G N M . . . A . W . E D C . . 1 0 7 6 . 4 8
28 S T U V O P Q R K L M N . H . J . . E . . . A B 4 5 6 . 0 W 2 . .
29 T S V U P O R Q L K N M H . . I D C F E . 8 B A 5 4 7 6 1 0 3 W 2
30 U V . T Q R O . . N K L I J . H E F C D A B 8 . 6 W . 5 . 3 0 1 7
31 V U T S R Q . O N . L K J I . G F E D C . A . . 7 6 5 W 3 . 1 0 .
32 0 . 3 G 6 L 5 M C V F S A P . . O B . 8 . D T E K 7 N 4 I 1 H 2 W
```

Provenance. The attached terminal output identifies array 29909858, task 0, rank 5. It was truncated with `cut -c-6000`, but both 1089-entry arrays are complete. Restoring the closing JSON brace recovers the grids; 844 is counted directly. No missing seed, elapsed time, iteration or search certification label is inferred. Independent comparison confirms the same 875-clue source completion and exactly 31 removed clues.

The normalized grids, raw attachment, two complete sets of 844 deletion witnesses and independent audit are in `lcs33_844_bunya_29909858/`. The earlier 841 certificate remains in its original directory. The square has not changed; the improved choice of clues produced the gain.

Historical verification: 841 at order 33

This 16 October milestone is retained; the latest verified size is 872 (page 2). The supplied record reports size 841, $t_2 = 380$, and search certification. Those labels were not assumed: all grid, uniqueness, essentiality and source-subset checks were rerun independently.

Quantity	Independently checked value
Clues / holes	841 / 248
Clues removed from the 875-input	34
Full-completion intercalates	6976
Private intercalates	488
Distinct clues with a private intercalate, t_2	380
Replayed essential-clue witnesses	841

Uniqueness and minimality. Local Boolean exact-cover and integer AllDifferent encodings each returned infeasible after excluding the stated completion (5.03 and 3.59 seconds respectively). A separate agent's independently written integer checker returned infeasible in 4.24 seconds. It also verified the Latin square, every clue's agreement, exact witness-key coverage, and all 841 alternate Latin squares. Each alternate changes its deleted clue while retaining all other clues. Finally, the audit verifies the original 875-source hash, identical completion, subset relation and exact 34-clue removal list. Uniqueness is solver-dependent; essentiality witnesses are portable. No optimality claim or portable UNSAT proof is made.

The fixed data are `n33_841.json` and `n33_841.in` in `lcs33_bunya_29909858/`. Run `python3 replay_record.py original_record.json` there for solver-free essentiality replay, or add `-check-unique -seconds 120` for a fresh integer uniqueness check. The separate agent's `verify.py` is in `independent_audit/`; it also checks source provenance.

Consequences.

$$\boxed{841 \le \mathrm{lcs}(33) \le 977}, \qquad 841 - \lceil E(33) \rceil = 7, \qquad 841 - 838 = 3.$$

The previous ceiling conjecture is exceeded by seven. The former sharp-plus lower target is surpassed at this order, and the prediction of equality with 838 is false. No conclusion about every $2^m + 1$ or about powers-of-two exactness follows solely from this one example.

Coherent construction profile. With $(s, t_2) = (841, 380)$ and only the safe statement $t_3 \ge 0$, the audited lift formulas give

$$H_2(33) = 3151, \qquad H_3(33) = 2 \cdot 1089 + 5 \cdot 841 = 6383, \qquad H_4(33) = 12426.$$

Thus $\mathrm{lcs}(99) \ge 6383$, improving 6358 by 25. The old 768-clue seed remains stronger for the binary and quaternary lifts: the best bounds at 66 and 132 remain 3383 and 14475. Each seed's size and witness count stay paired.

Verified milestones: 447 at order 27 and 844 at order 33

Both candidates were checked without relying on their search certification. Boolean exact-cover and integer AllDifferent models independently exclude an alternative completion and produce a witness for every removed clue. All saved witnesses replay directly as Latin squares. A separate agent rechecked uniqueness, witnesses, structural counts and source comparisons.

Quantity	Order 27	Order 33
Clues / holes	447 / 282	844 / 245
Previous verified size	446	841
Clues with a private intercalate, t_2	110	367
Clues with an order-three witness, t_3	22	0
Completion intercalates	648	6976
Order-three subsquares	81	0
Boolean uniqueness, seconds	0.09	1.96
Integer uniqueness, seconds	0.07	2.35
Separate agent uniqueness, seconds	0.09	2.01
Deletion witnesses replayed, two sets	$2 \cdot 447$	$2 \cdot 844$

Order 27. Array 29909858, task 17, rank 6, seed 299098580142, square `lab_special27-conjugate`, strategy `chunk-shuffle`, 6873 iterations, reported elapsed time 25462.17 seconds. The completion is identical to that of the 446-set. The new set retains 442 clues, removes four and adds five. Its first row is empty. The gap to the conjectural 508 target is now 61; that target is not an optimum certificate.

There is an exact product description. Put $B_{r,c} = L_{r,c}$ for $0 \le r, c < 9$, and

$$A = \begin{pmatrix} 0 & 1 & 2 \\ 2 & 0 & 1 \\ 1 & 2 & 0 \end{pmatrix}. \qquad L_{9a+r,9b+c} = 9A_{a,b} + B_{r,c}.$$

Both A and B are Latin. This describes the square, but does not by itself supply a general theorem for critical-set sizes. The verified set gives $447 \le \mathrm{lcs}(27) \le 638$ under the report's audited upper bound.

Order 33. The attached output was truncated after the complete square and partial arrays; size 844 was counted locally. The claimed search status, seed and timing were unavailable. All proof checks were therefore performed from the grids. This is a 31-clue deletion from the same 875-input, with the same completion as the earlier 841 example. Thus

$$\boxed{844 \le \mathrm{lcs}(33) \le 977}, \qquad 844 - \lceil E(33) \rceil = 10, \qquad 844 - 838 = 6.$$

The ceiling conjecture remains false, now exceeded by ten. Uniqueness is solver-dependent; no portable UNSAT proof or optimality claim is made.

Protected lifts. The 447 profile gives $(H_2, H_3, H_4) = (1733, 3737, 7104)$. Its ternary score improves the 446 profile by seven but does not beat the existing bound 4503 at order 81. The 844 profile gives $(H_2, H_3, H_4) = (3144, 6398, 12357)$, hence

$$\boxed{\mathrm{lcs}(99) \ge 6398}, \quad \text{up by 15 from 6383.}$$

Older profiles remain in the envelope when stronger; size and witness counts from different seeds are never combined.

Reproduction. The directories `lcs27_bunya_29909858/` and `lcs33_844_bunya_29909858/` retain inputs, raw provenance and both certificate sets. Run `python3 check_certificates.py n27` or `n33` respectively for solver-free witness replay. Their separate `independent_audit/` directories contain the independent checks.

Historical campaign 29909858: at most 192 CPUs

The completed allocation below produced verified 844 at order 33, surpassing its 838 target. It is preserved as historical experimental provenance. The successor array 30001580 is reported on page 2; its five new exported records have now been independently verified. Success at this one order does not prove the lower family for every m. The remaining lower-envelope targets are experiments, not certified maxima.

Focus	Jobs	CPUs	Experiment and success criterion
Order 33	12	96	Four jobs repair the 875-input in randomized orders; four grow and repair the 836/831 sets; four explore nonlinear XOR prolongations, traded XOR prolongations, traded special33, and a random-square control. 838 surpassed; seek 845 and beyond.
Order 30	3	24	Two jobs grow the 654-set; one explores different squares. Seek 663.
Order 28	2	16	Grow the 536-set and explore different squares. Seek 559.
Order 27	2	16	Grow the 446-set and explore different squares. Seek 508.
Orders 9, 15	2	16	One job per order. Improve 47 or 151; sizes 50 or 153 would give additional counterexamples to the ceiling upper conjecture.
Product seeds	3	24	One job each at orders 14, 15, 33, maximizing the proved lift scores rather than size alone.
Total	24	192	24 hours/job; at most 4608 allocated CPU-hours.

Why change the search. The former oracle left the decisive 875-input unresolved. The follow-up uses the forced-change CP-SAT oracle, retains alternate-completion witnesses, and adds randomly selected clues before pruning an elite set. Additions let the search leave subsets of the original 875-mask. Fresh squares start fully filled, so their initial uniqueness is immediate. Randomized construction sampling is not claimed uniform. Unresolved candidates are checkpointed separately; unknown checks never count as certification. New outputs remain *search-certified pending independent verification*, even though every recorded result has a fresh uniqueness check and a replayable witness for every clue.

Product objectives. At order 14 maximize $2s + t_2$: a score of at least 363 gives $\mathrm{lcs}(28) \geq 559$. At order 15 a score of at least 438 gives $\mathrm{lcs}(30) \geq 663$. The order-33 product job splits its eight workers between $2s + t_2$ and $3s + 6t_2$, aiming to improve the safe 3383 and 14475 bounds at orders 66 and 132. These targets use coherent size/witness pairs and do not assume an unrestricted product theorem.

Execution. The package `lcs_hpc_followup_192/` contains 24 array tasks with eight independent single-CPU workers each, a 24-hour wall limit, CP-SAT configured for one worker per process, and resumable checkpoints. For a fresh submission, run the supplied ten-minute smoke test before the full array. This campaign has finished; array 29909858 is the source of 151 at order 15, 447 at order 27 and 844 at order 33. The package was tested locally and its startup checks corrected after an early launch failure. No success probability or 24-hour guarantee is asserted.